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Ramkumar et al.

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(54) **MANAGED ACTIVATION OF LICENSED SOFTWARE FOR RESOURCES IN PEERED NETWORKS**

(56) **References Cited**
U.S. PATENT DOCUMENTS

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8,363,658 B1 * 1/2013 Delker H04L 41/0895 370/252
11,240,203 B1 * 2/2022 Eyada H04L 63/20
2013/0239232 A1 * 9/2013 Yoakum G06F 21/121 726/29

(Continued)

FOREIGN PATENT DOCUMENTS

AU 2006275655 B2 * 2/2007 G06F 21/105
CN 106534084 A * 3/2017 G06F 21/105

(Continued)

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G06F 21/10 (2013.01)
H04L 9/40 (2022.01)

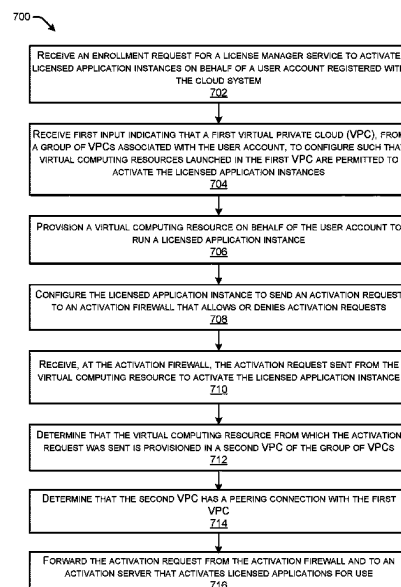
(52) **U.S. Cl.**
CPC **G06F 21/105** (2013.01); **H04L 63/0272** (2013.01); **H04L 63/102** (2013.01); **G06F 21/1075** (2023.08)

(58) **Field of Classification Search**
CPC G06F 21/105; G06F 21/1075; H04L 63/0272; H04L 63/102
See application file for complete search history.

(57) **ABSTRACT**

Techniques for a license manager service of a cloud system to activate and run licensed applications in VPCs with peering connections (or “peered VPCs”) established with configured VPCs. The license manager service may provide users with the ability to select a particular VPC to be configured to use licensed applications, and may further allow users to enable peered VPCs of the configured VPC to use licensed applications. The license manager service may provide a managed experience for activating the licenses for the applications running on computing resources in the configured VPCs and peered VPCs. The license manager service may launch licensed applications on virtual resources in a user’s configured VPC and peered VPCs, manage the process for activating the licensed applications with third-party providers, and provide the user with access to licensed applications that have been activated and configured by the license manager service in configured and peered VPCs.

20 Claims, 12 Drawing Sheets



(56)

References Cited

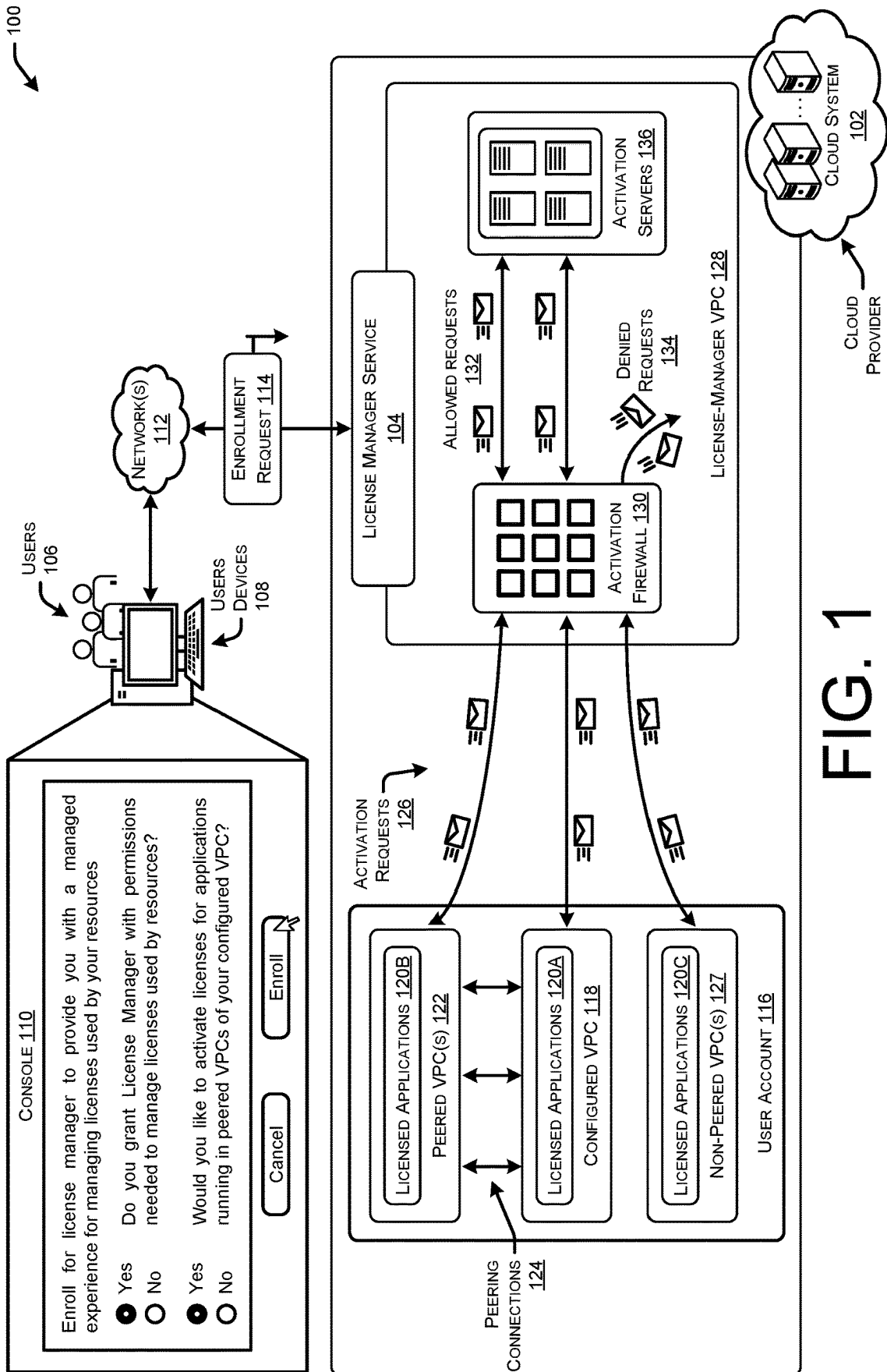
U.S. PATENT DOCUMENTS

2018/0375815 A1 * 12/2018 Dinardo, Sr. H04W 4/025
2020/0344229 A1 * 10/2020 Srivastava G06F 21/105
2023/0125189 A1 * 4/2023 Chang H04L 63/12
726/26

FOREIGN PATENT DOCUMENTS

CN 109255208 A * 1/2019 G06F 21/105
EP 4446912 A1 * 10/2024 G06F 21/1014
WO WO-2008115647 A1 * 9/2008 G06F 21/105
WO WO-2015087444 A1 * 6/2015 G06F 21/10
WO WO-2023069392 A1 * 4/2023 H04L 12/46
WO WO-2024238543 A1 * 11/2024

* cited by examiner



200

LICENSE MANAGER ENROLLMENT

Enroll for license manager to provide you with a managed experience for managing licenses used by your resources

202 **Enable License Manager**

Create a service-linked role for managing subscriptions -

Do you give License Manager permission to create a service-linked role in your account to manage licenses used by resources?

☒ I agree to give License Manager permission to create a service-linked role in my account

204 **Configure Directory Service**

Managed Directory Service

206 Choose the directory service and products you want to have managed

Directory name and ID

208

Product names and IDs

210

Virtual Private Clouds

Choose a VPC for additional configuration to enable License Manager

VPC ID	Status
☒ VPC-e29b343	☐ Unconfigured
☐ VPC-e31d232	☐ Unconfigured

214 ☒ Would you like to activate licenses for applications running in peered VPCs of your configured VPC?

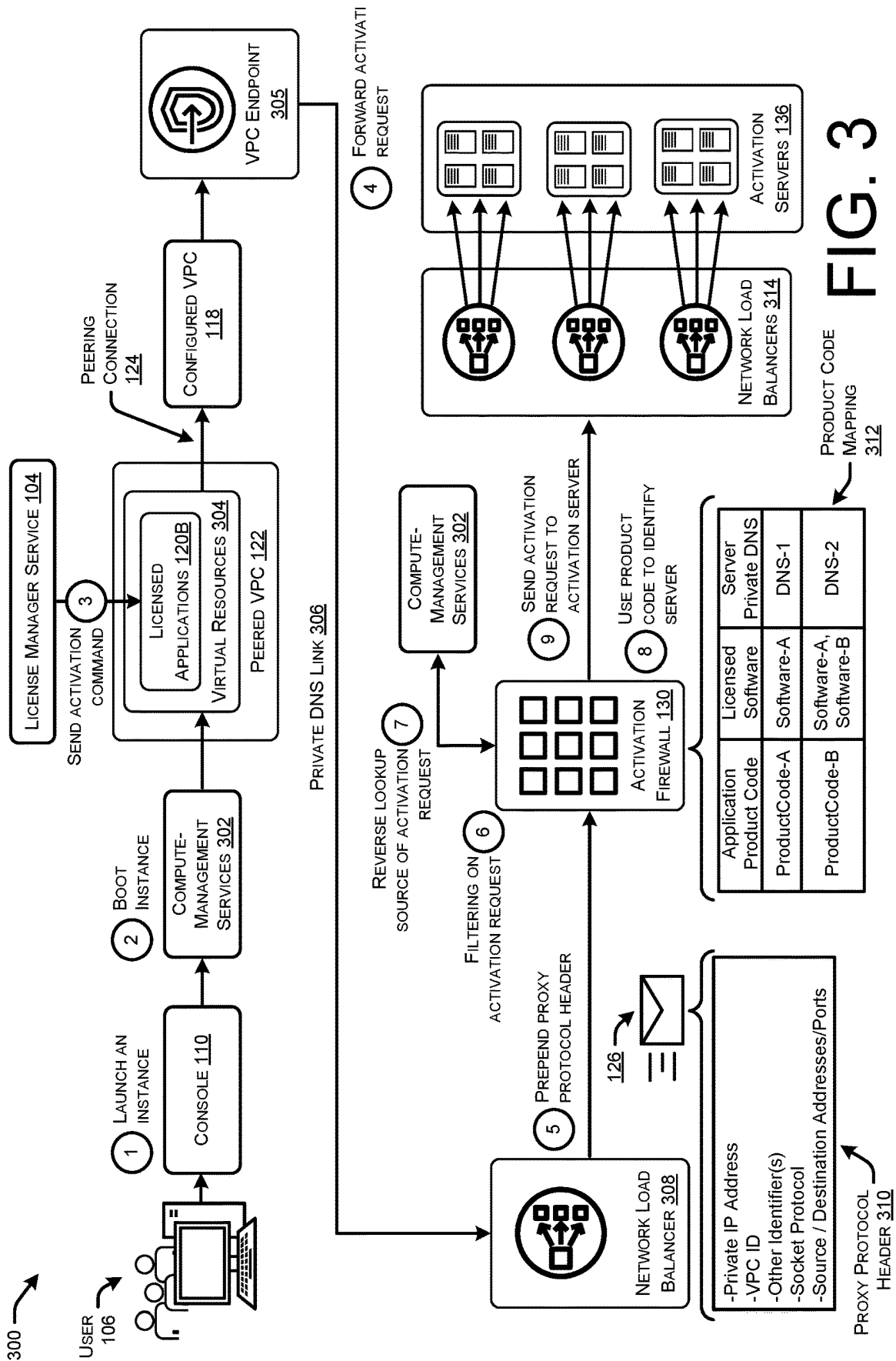
216 **Subnets for VPC-e29b343**

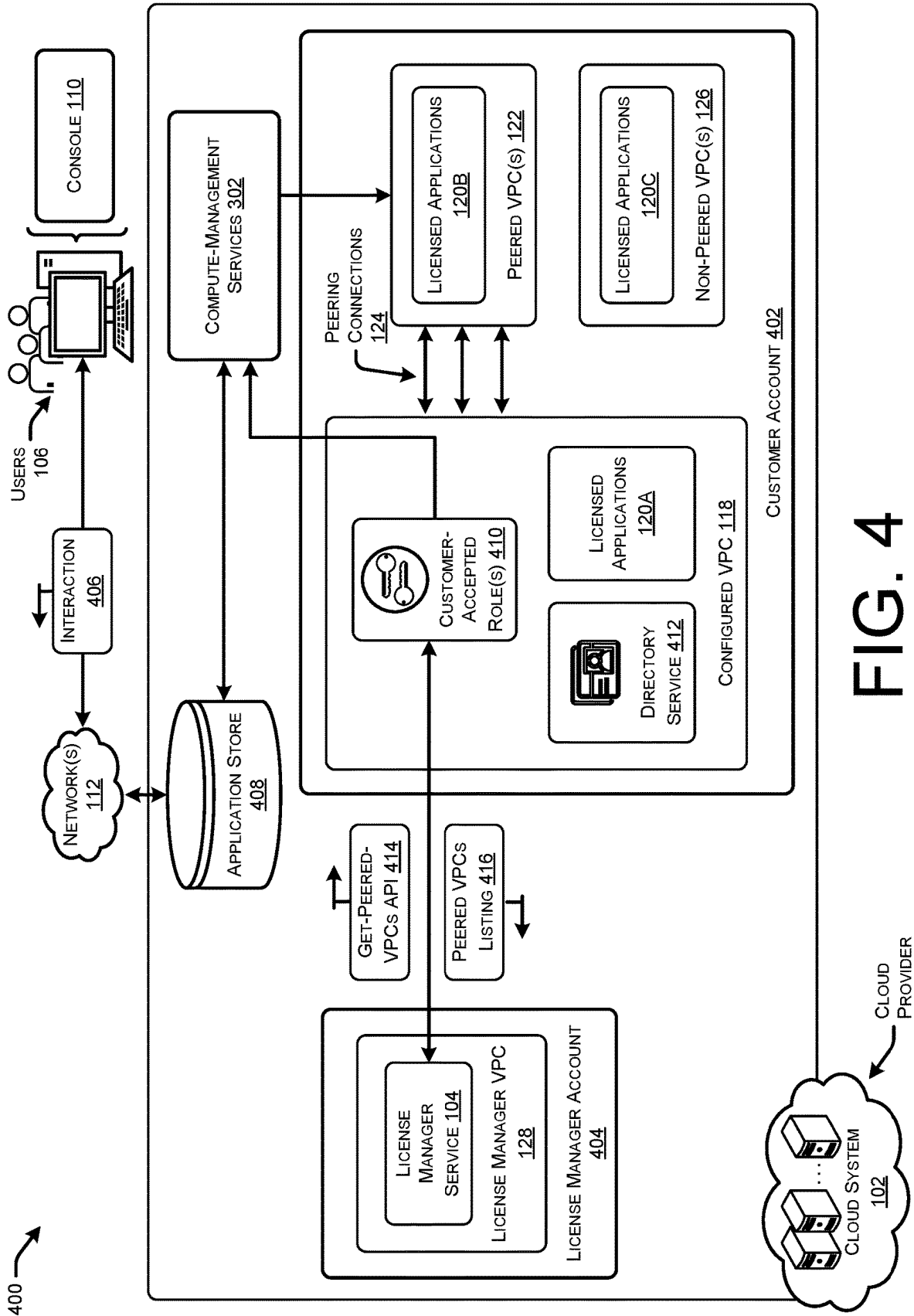
Select a subnet in which to configure a VPC endpoint

218 **Security groups for VPC-e29b343**

Select a security group that allows traffic inbound to the VPC endpoint

FIG. 2





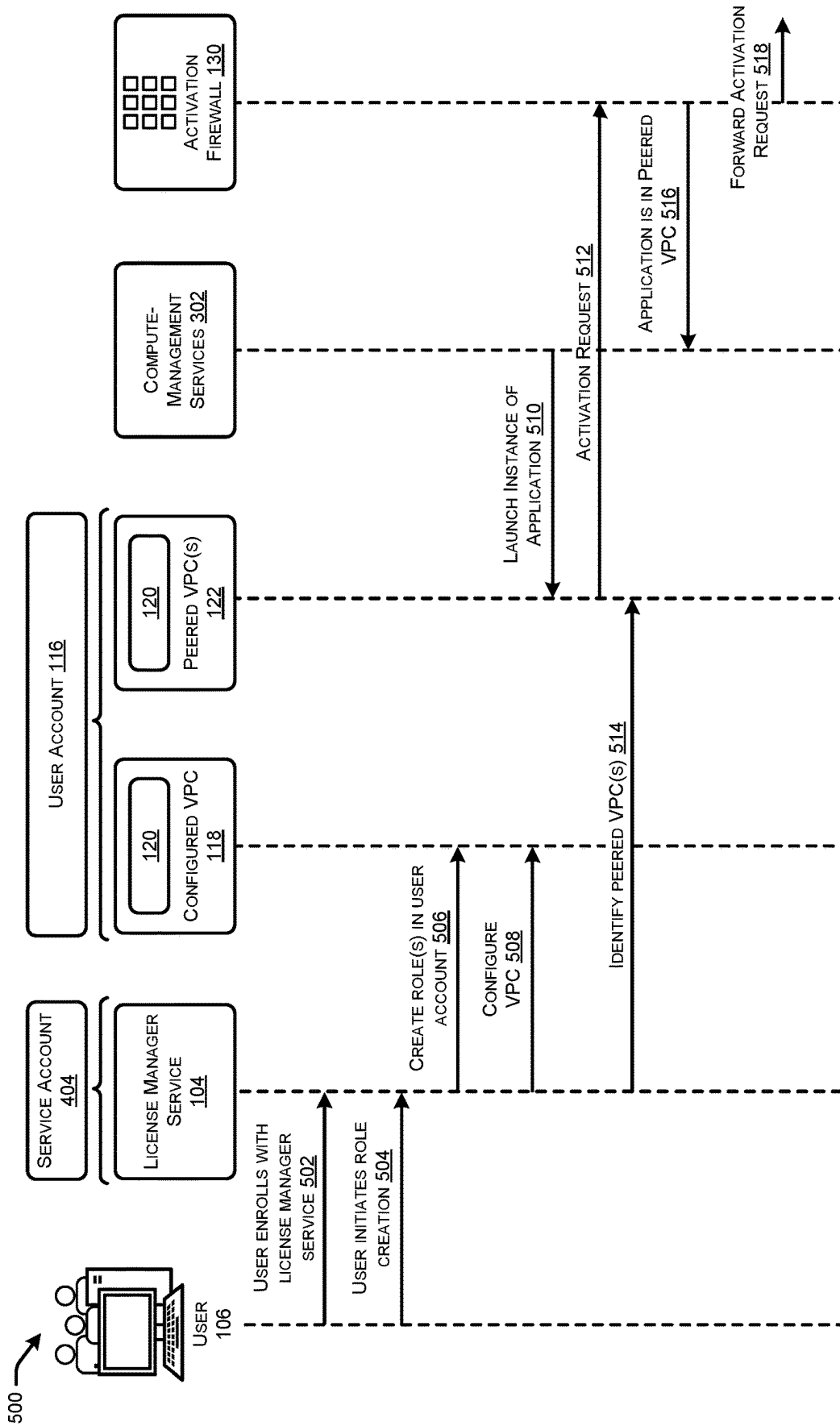


FIG. 5

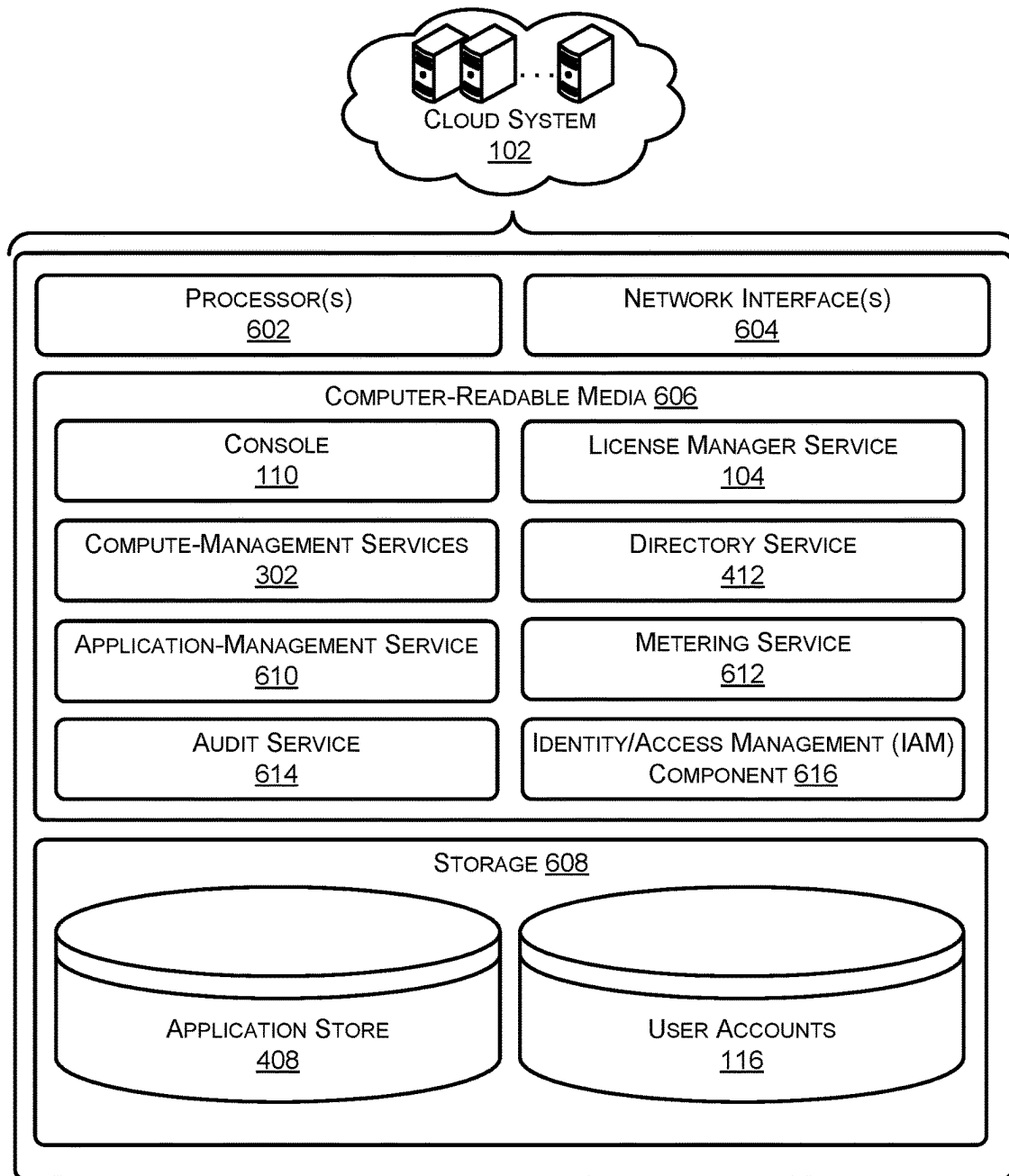
600 

FIG. 6

700

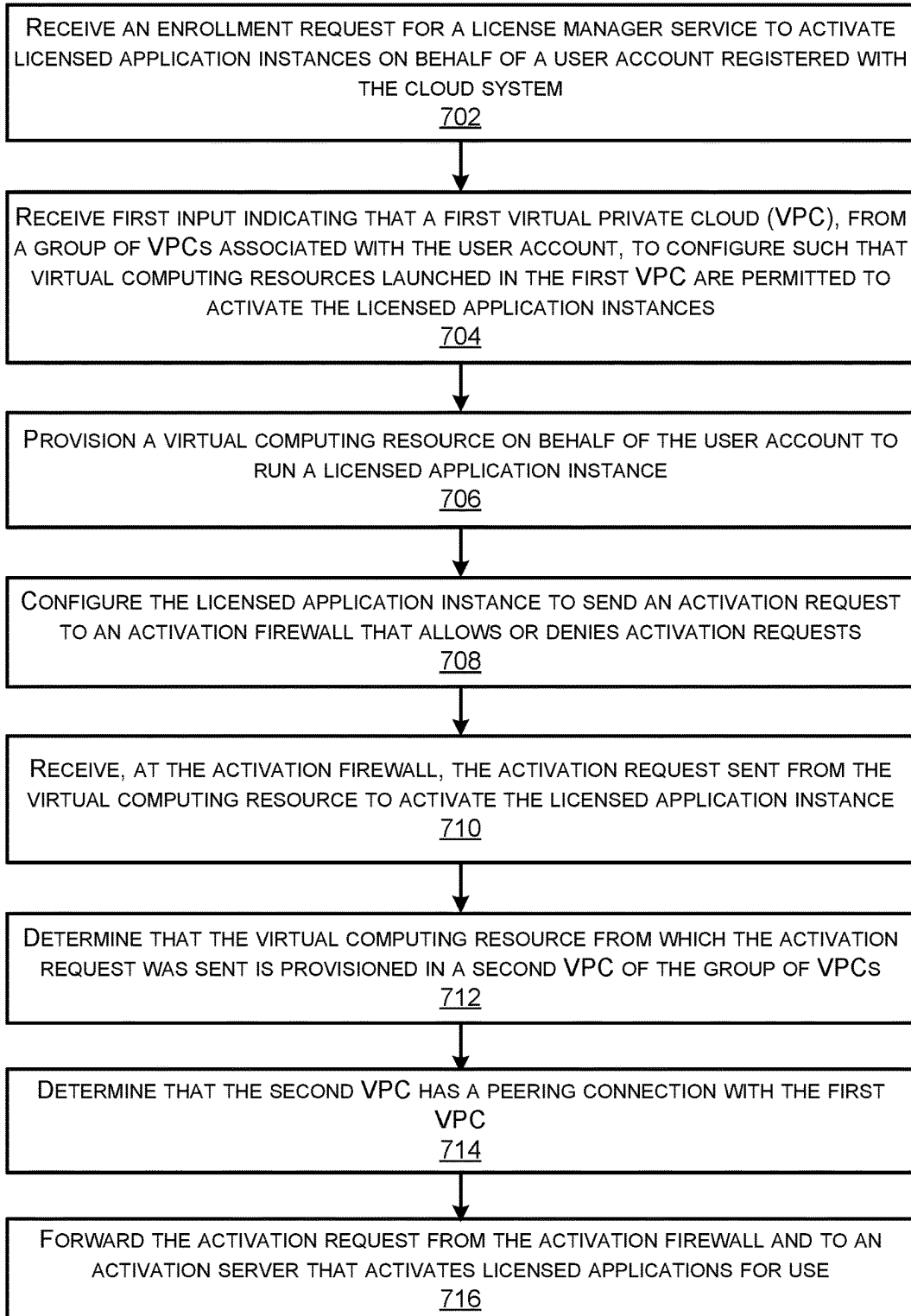


FIG. 7

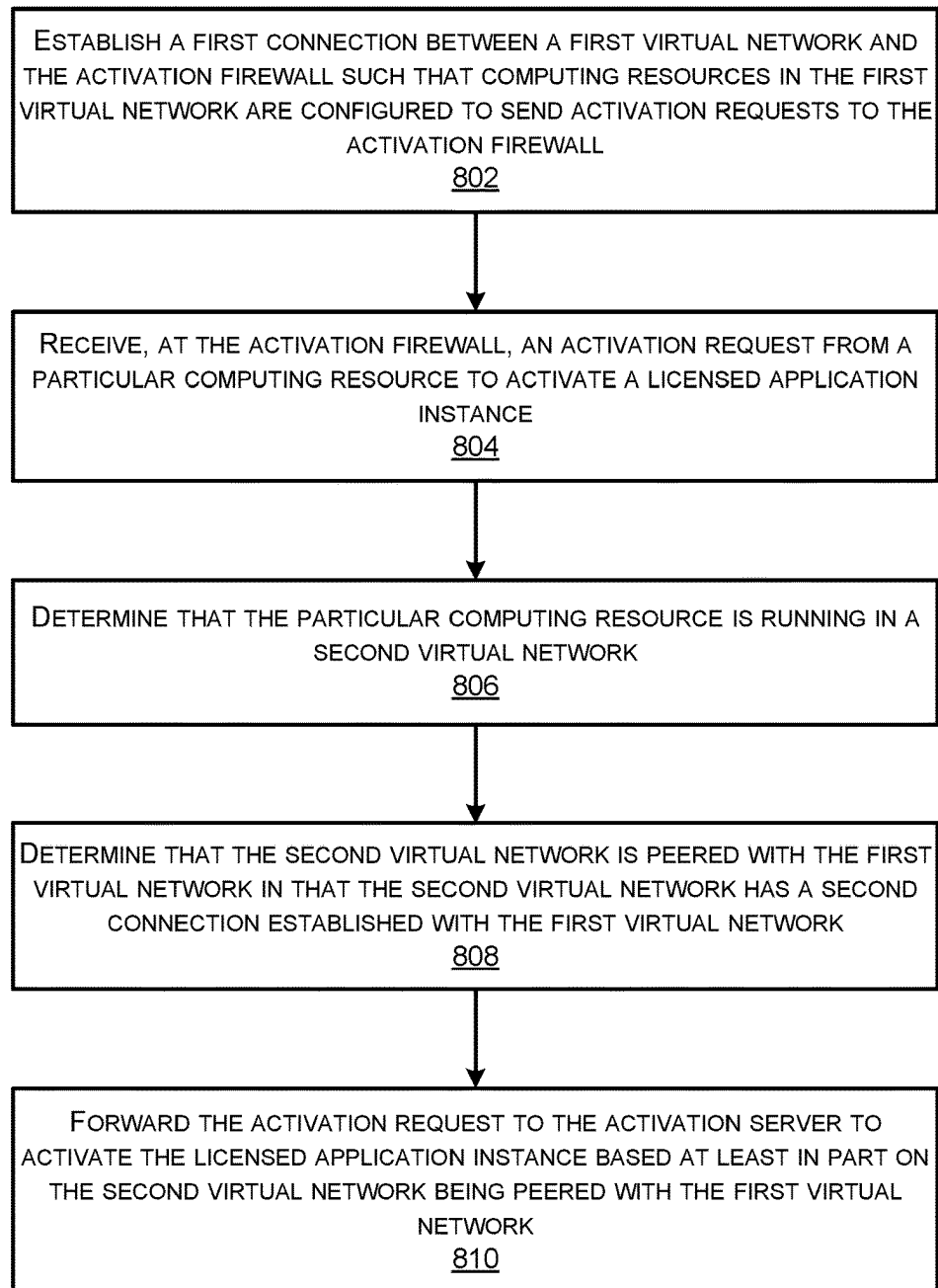
800 

FIG. 8

900 ↗

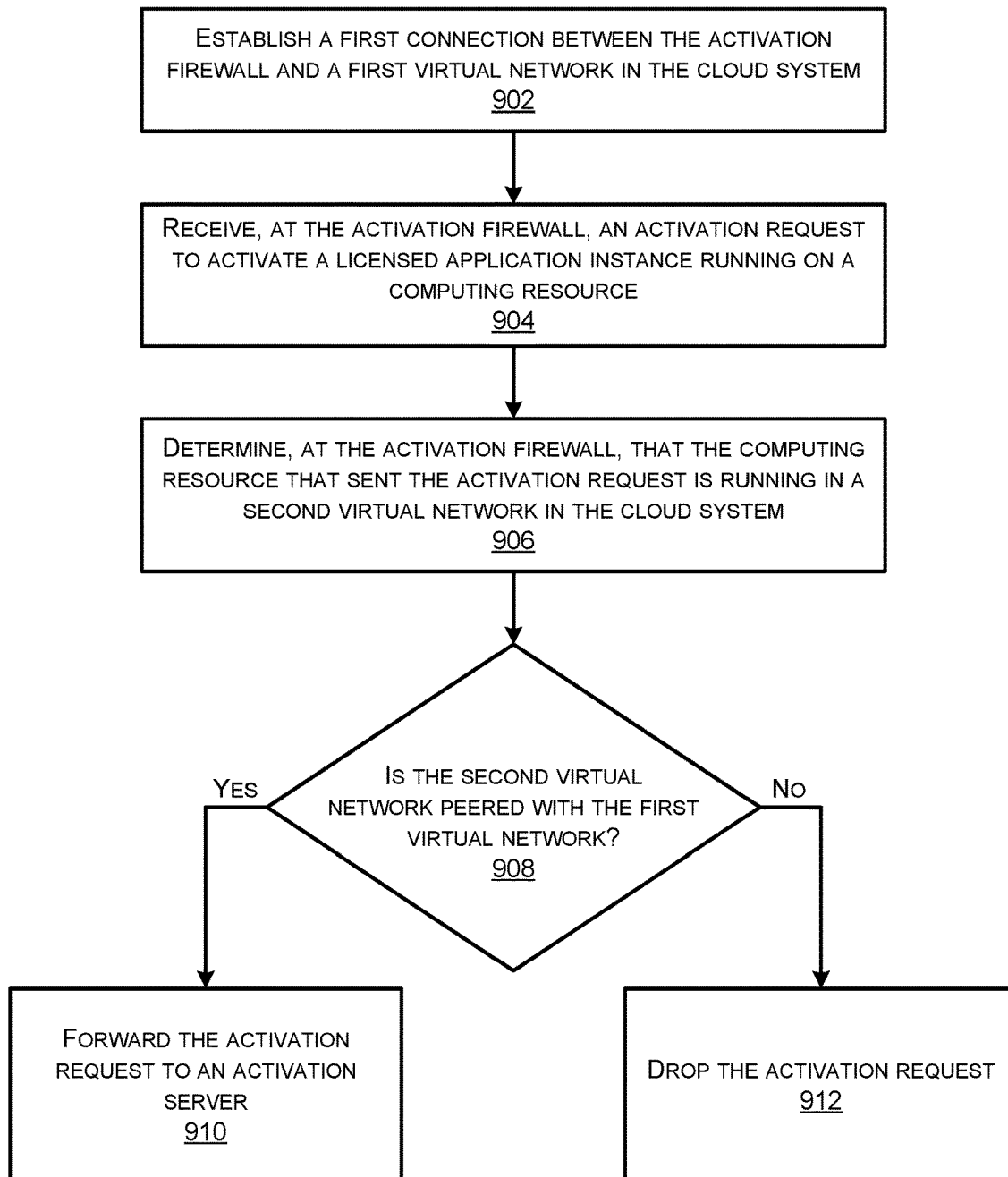


FIG. 9

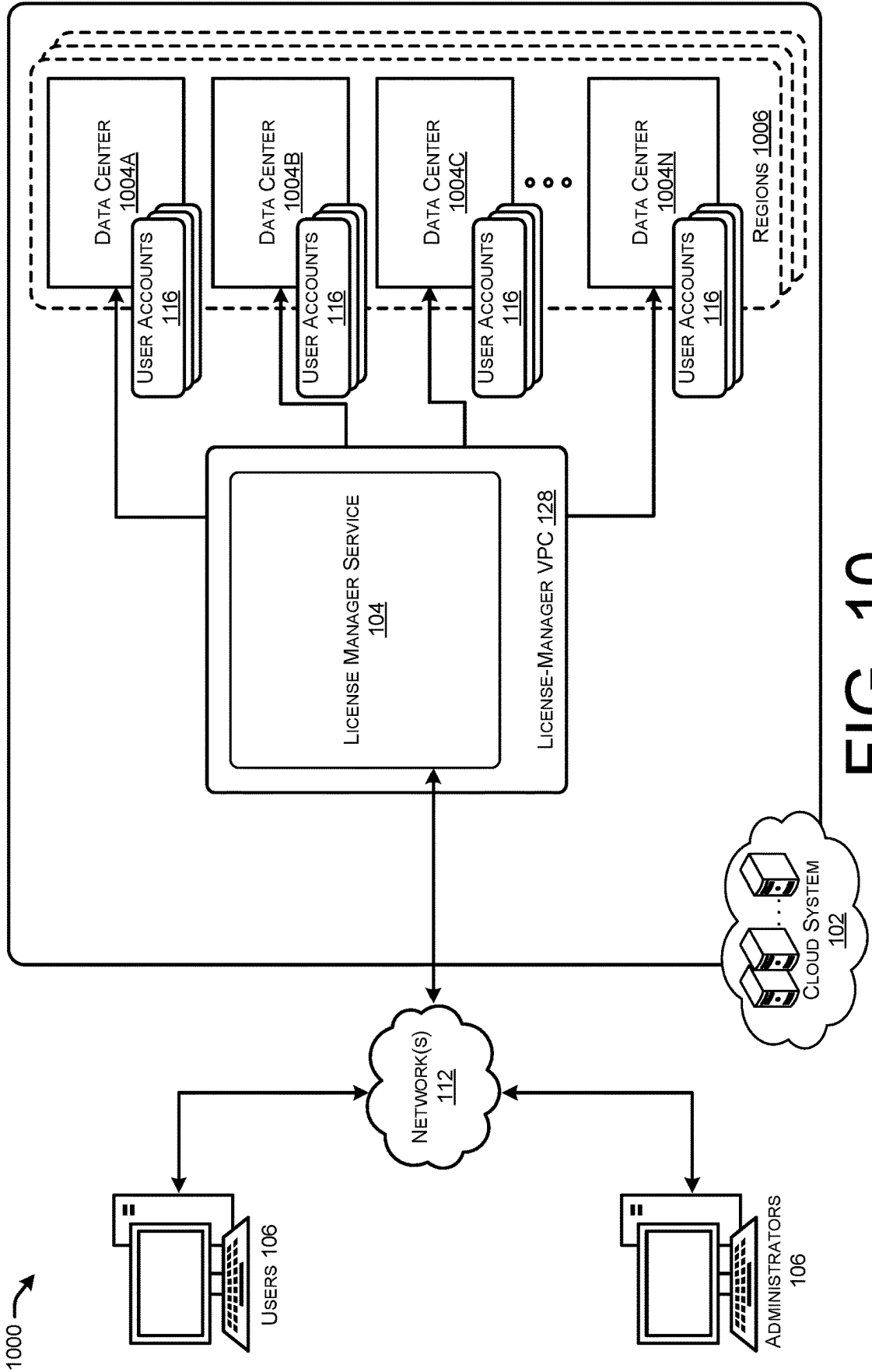
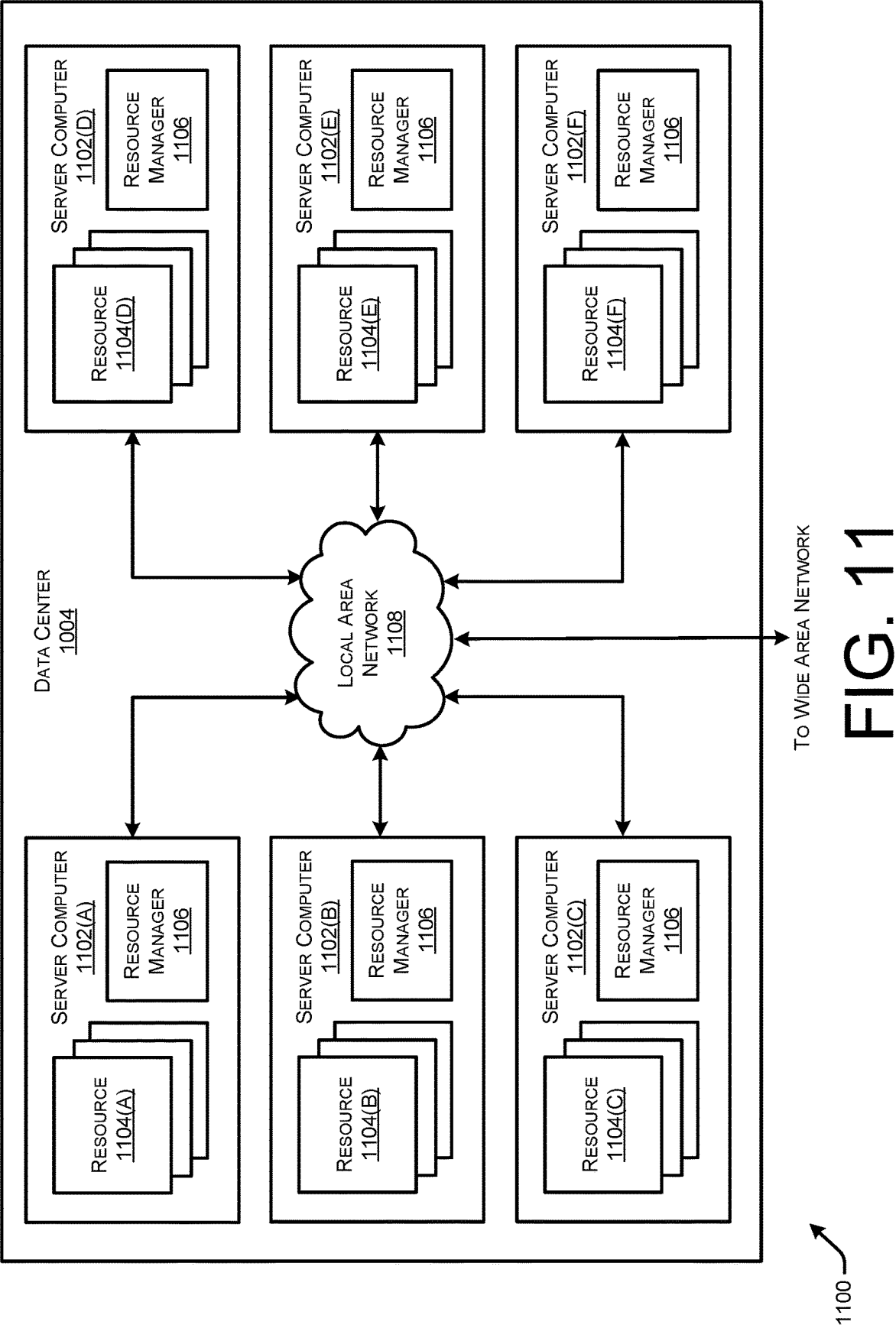
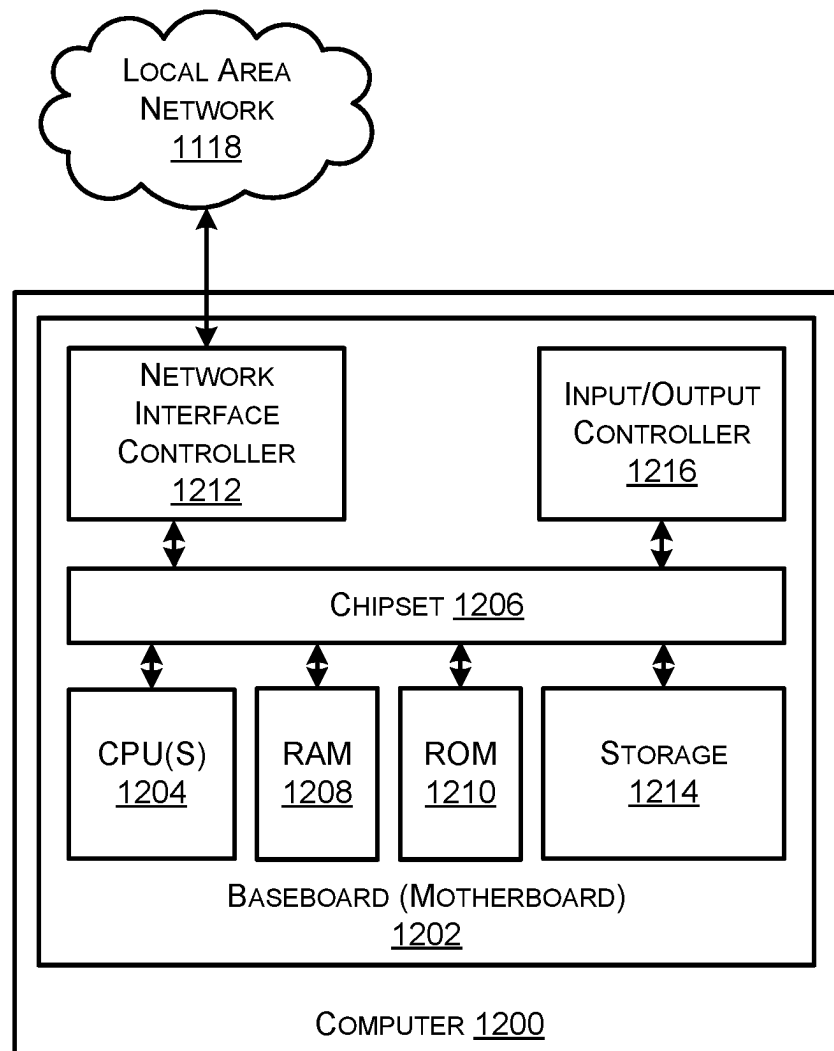


FIG. 10



**FIG. 12**

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MANAGED ACTIVATION OF LICENSED SOFTWARE FOR RESOURCES IN PEERED NETWORKS

BACKGROUND

Cloud providers offer cloud-based services to fulfill users' computing-service needs without the users having to invest in and maintain computing infrastructure required to implement their services. These cloud providers maintain networks of managed computing resources and functionality to implement various types of scalable, on-demand services, to support applications of the users. Specifically, the cloud providers may maintain networks of data centers, which in turn may include a number of interconnected computing devices (or "servers"), that provide computing resources to support applications of the users. Users may be provided with logically isolated portions of the cloud provider network, often referred to as virtual networks (VNETs) or virtual private clouds (VPCs), in which they can launch resources to support their applications. Users of cloud systems can launch and run various types of applications on scalable resources in their VPCs, such as their own proprietary applications, cloud-native applications provided by the cloud provider, and third-party applications provided by third-party developers.

BRIEF DESCRIPTION OF THE DRAWINGS

The detailed description is set forth below with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items. The systems depicted in the accompanying figures are not to scale and components within the figures may be depicted not to scale with each other.

FIG. 1 illustrates a system-architecture diagram of an example environment in which an activation firewall of a license manager service allows activation requests from licensed applications in peered VPCs to be sent to activation servers.

FIG. 2 illustrates an example user interface through which a user enrolls for use of a license manager service.

FIG. 3 illustrates a flow diagram in which a license manager service and components of a cloud system activate a licensed application running in a peered VPC for use by a user account.

FIG. 4 illustrates a system-architecture diagram of an example environment in which a license manager service uses roles in a user account to determine a listing of peered VPCs for a configured VPC in the user account.

FIG. 5 illustrates a flow diagram of an example environment in which a license manager service activates a licensed application in a peered VPC of a configured VPC.

FIG. 6 illustrates a component diagram of example components of a cloud system that performs the techniques described herein.

FIG. 7 illustrate a flow diagram of an example process for a license manager service to configure a VPC, determine peered VPCs for the configured VPC, and forward an activation request sent from a licensed application in the peered VPC to an activation server.

FIG. 8 illustrate a flow diagram of an example process for a license manager service to receive an activation request sent from a licensed application, determine that the licensed

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application is in a peered VPC of a configured VPC, and forward the activation request to an activation server.

FIG. 9 illustrate a flow diagram of an example process for an activation firewall to filter activation requests from licensed applications.

FIG. 10 is a system and network diagram that shows an illustrative operating environment that includes data centers of a cloud system that can be configured to implement aspects of the functionality described herein.

FIG. 11 is a computing system diagram illustrating a configuration for a data center that can be utilized to implement aspects of the technologies disclosed herein.

FIG. 12 is a computer architecture diagram showing an illustrative computer hardware architecture for implementing a computing device that can be utilized to implement aspects of the various technologies presented herein.

DETAILED DESCRIPTION

This disclosure describes techniques for a license manager service of a cloud system to activate and run licensed applications in VPCs that have peering connections (or "peered VPCs") established with configured VPCs of the users. Manually tracking the usage of software licenses for licensed applications can be cumbersome and error-prone. Further, the process of activating the licenses for third-party applications (e.g., applications developed by entities other than the cloud provider and the users) may be difficult or impossible depending on the activation protocols and procedures put in place by the application providers. The license manager service may provide users with the ability to select a particular VPC to be configured to use licensed applications, and may further allow users to enable peered VPCs of the configured VPC to use licensed applications. The license manager service may provide a managed experience for activating the licenses for the applications running on computing resources in the configured VPCs and peered VPCs. The license manager service may launch licensed applications on virtual resources in a user's configured VPC and peered VPCs, manage the process for activating the licensed applications with third-party providers, and provide the user with access to licensed applications that have been activated and configured by the license manager service in configured and peered VPCs.

Because VPC constructs provide isolation and security to users by preventing others from accessing their VPCs, the users generally have to personally manage various aspects around the resources and applications in their VPCs. However, with such a wide variety of applications running in the cloud system, it can take a great deal of time and expertise for users to launch, configure, and manage applications running in their VPCs. According to the techniques described herein, users may enroll for more managed and automated experiences than those traditionally provided by managed services, such as license manager. For example, users may request that license manager service perform different types of automated or semi-automated functions such as launching, configuring, licensing, and activating licenses for various licensed applications for computing resources (e.g., servers, VMs, containers, etc.) in configured and peered VPCs. To opt in for the functionality provided by the license manager, users may define or provide a set of permissions, or roles, that the license manager service may assume while in the user account. The permissions define types of management operations that the license manager service is permitted to take, and/or not permitted to take, as the license manager service provides various automated

management experiences. In some instances, the set of permissions created for the license manager service may be referred to as a “service-linked role” that the managed service may assume when performing management operations in the user account.

After the set of permissions are in the user account, the license manager service may request that the user designate a VPC to be configured for the licensed applications to be launched into, and to communicate with activation servers. Generally, the cloud system may include one or more activation servers that perform techniques for activating licenses of the licensed applications. The VPC that is selected for configuration will include at least one sub network (also referred to herein as “subnet”) in which an endpoint (or “gateway”) is provisioned that is configured to create a private connection with the activation servers over which activation requests are sent to activate the licensed applications. The license manager service may configure the selected VPC by launching a VPC endpoint into a subnet of the VPC to establish the private connection with the activation server(s), applying security groups or policies that allow communications with the activation servers, and configuring a directory service in the VPC that manages permissions and credentials for activating the licensed applications. In some examples, virtual computing resources on which licensed applications are launched in the configured VPC may be allowed to activate the licensed applications.

The user account may have additional VPCs in which computing resources are hosted, and some of those VPCs may have peering connections with the configured VPC. Generally, VPCs are overlay networks that are logically isolated from other overlay networks in the cloud system in which computing resources are launched. A VPC peering connection is a networking connection between two VPCs that enables traffic to be routed between the VPCs using private IP addresses (e.g., IPv4 and/or IPv6 addresses). Computing resources in peered VPCs can communicate with each other over the peering connections as if they are within the same virtual network. The VPC peering connections can be created between VPCs registered to a single user account, or between VPCs in different user accounts, and the VPCs can be in different regions (also known as an inter-region VPC peering connection).

According to the techniques described herein, the user account may further specify that the license manager service is to activate licensed applications that are launched in peered VPCs of the configured VPC. The peered VPCs may communicate the activation requests from the licensed applications running in the peered VPCs over the peered connection to the configured VPC, and the activation requests may then be forwarded to an activation server over the private connection between the configured VPC and the activation server.

As an example, the user account may request that the license manager, or another service of the cloud system, provision a new instance of a licensed application in a peered VPC of the configured VPC. The license manager service may launch a virtual resource (e.g., VM, container, etc.), or detect a launch of a virtual resource, on which the licensed application is installed. The license manager service may configure the licensed application to send activation requests to an activation firewall that is configured to allow or deny activation requests destined to the activation server. For instance, the licensed application may be configured to send activation requests to a private Domain Name System (DNS) name that is assigned to the activation firewall, and/or a load balancer that distributes activation

requests to the activation firewall. To trigger the activation process, the license manager service may send a command to the licensed application to send the activation request to the activation firewall and begin the activation process. The licensed application running in the peered VPC may send the activation request over the peering connection to the configured VPC, and the activation request may then be forwarded to the activation firewall over the private connection.

The activation firewall may be used to ensure that licensed applications are only activated if activation of the licensed applications are permitted by the license manager. In this way, the license manager service may track usage of all licensed applications and determine metrics indicating the usage of the licensed applications for compliance reasons. For instance, there may be agreements in place regarding usage, such as payment or permitted use agreements, that indicate amounts of usage of licensed applications that are permitted, or costs for running licensed applications (e.g., cost per instance of a licensed application, cost for how long a licensed application is used, etc.).

To ensure that users are unable to provision and activate license instances that are not tracked by the license manager, the activation firewall may filter activation requests received from virtual computing resources. Following the above example, the activation firewall may receive the activation request sent from the application on the virtual computing resource and determine the source of the activation request. For instance, the activation firewall (and/or load balancer) may extract metadata from the activation request indicating that the virtual computing resource is a source of the activation request. The metadata may include a private Internet Protocol (IP) address associated with virtual computing resource, a VPC ID of a VPC in which the virtual computing resource is provisioned, and an account identifier (ID) of the user account. The activation firewall may send at least a tuple including the VPC ID of the peered VPC and the private IP address (and potentially the user account ID) of the virtual computing resource to the license manager service and/or a compute-management system to verify that the virtual computing resource is in a peered VPC to the configured VPC of the user account.

The activation firewall may be configured to drop, block, or otherwise refrain from sending activation requests to the activation server(s) upon determining that the activation requests were received from virtual computing resources and/or applications that are not running in the configured or peered VPCs. Conversely, the activation firewall may forward activation requests to the activation server upon determining that the activation requests were received from virtual computing resources and/or applications that are running in the configured or peered VPCs. In this way, only licensed applications managed by the license manager service and launched in the configured or peered VPCs may be activated by activation servers.

In some instances, there may be different activation servers configured to activate different licensed applications. The activation firewall may determine a unique product code for the licensed applications. The activation firewall may store mappings between private DNS names for activation servers and unique product codes, and forward activation requests to the private DNS names that are mapped to the requests’ unique product codes. In some instances, the activation firewall and activation servers may be in a service VPC of the cloud system. In other instances, however, the activation servers may be at a remote location outside of the cloud system.

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The activation firewall may help establish and maintain a connection open between the virtual computing resource and the activation server to allow for activation of the application using an activation protocol. The connection may be a stateful Transport Control Protocol (TCP) connection that is used to communicate data for the activation protocol. In this way, a license manager service may automate the process for provisioning and activating licensed applications for use by users of a cloud system.

In some examples, the cloud system may offer customer accounts that allow organizational users (e.g., enterprises, companies, customers, groups of users, etc.) to create user accounts that are managed by administrators of the customer account. For instance, companies may register for one or more customer accounts under which individual user accounts are created for employees of the companies. The customer accounts allow for administrators of those accounts to allocate resources, group accounts, and apply governance policies to accounts or groups.

In addition to provisioning and activating licensed applications, the license manager service may additionally assign user accounts of the customer account to have access to applications (licensed and unlicensed applications). To provide user accounts with access to licensed applications, the customer account may onboard a directory service with the manager service. For instance, an administrator of the customer account may provide the manager service with a directory ID of the customer account, and the manager service may locate and onboard the directory service using the directory ID. The administrator may provide input indicate what user account(s) in the customer account are to be provided access to use the licensed application. The license manager service may then, using the set of permissions, update the directory service to indicate the user account(s) that are permitted to access the application running in the customer account of the particular type. In this way, the applications running on virtual resources are usable by the assigned user accounts when the accounts present credentials (e.g., username and password, bio authenticator, etc.).

The techniques described herein solve problems rooted in cloud computing technology using solutions rooted in technology. Further, the techniques described herein improve the functioning of cloud systems. As an example, the process of activating licenses for third-party applications may be difficult or impossible depending on the protocols used and/or restrictions placed by the third-party developers. The techniques described herein enable the activation of these licensed applications such that cloud systems can support and run a greater variety of applications, thereby improving the functioning of the cloud system. Further, various cross-account management operations have traditionally been disallowed or were not previously possible. The techniques described herein enable additional cross-account management operations, which allows for cloud systems to offer managed services that can provide additional managed experiences for users.

Certain implementations and embodiments of the disclosure will now be described more fully below with reference to the accompanying figures, in which various aspects are shown. However, the various aspects may be implemented in many different forms and should not be construed as limited to the implementations set forth herein. The disclosure encompasses variations of the embodiments, as described herein. Like numbers refer to like elements throughout.

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FIG. 1 illustrates a system-architecture diagram of an example environment **100** that includes a cloud system **102** in which a license manager service **104** allows activation requests from licensed applications in peered VPCs to be sent to activation servers.

The cloud system **102** may be operated and/or managed by a cloud or service provider and may provide various services to users to fulfil their computing resource needs, such as cloud-based computing resources. For instance, users **106** may operate user devices **108** in order to register for use of the computing resources of the cloud system **102**. The cloud system **102** may include a one or more managed services that include components to provide different types of automated, or semi-automated, services for users, such as a license manager service **104**. Generally, the license manager service **104** may be, at least partly, control-plane systems that control operations occurring in the cloud system **102**. The license manager service **104** may be either centralized, or distributed, and be supported by one or more computing devices.

The license manager service **104** may integrate with other cloud services to simplify the management of licenses for software across user accounts, IT catalogs, and on-premises, through a single user account. The license manager service **104** lets users **106** (e.g., administrators) create customized licensing rules that emulate the terms of their licensing agreements, and then enforces these rules when a computing resource (e.g., server, virtual machine, container, etc.) gets launched. The rules in the license manager service **104** enable users **106** to limit a licensing breach by physically stopping the instance from launching or by notifying administrators about the infringement.

As illustrated, a cloud system **102** may be operated and/or managed by a cloud provider. The cloud system **102** may provide various services to users to fulfil their computing resource needs, such as cloud-based computing resources. For example, the cloud system **102** may provide cloud-based, scalable, and network accessible compute power services, storage services, database services, and/or other services. Users **106** may utilize user devices **108** to subscribe for use of the computing resources and/or services provided by the cloud system **102**.

A cloud system **102**, often referred to as a cloud provider network or simply as a “cloud,” refers to a pool of network-accessible computing resources (such as compute, storage, and networking resources, applications, and services), which may be virtualized or bare-metal. The cloud can provide convenient, on-demand network access to a shared pool of configurable computing resources that can be programmatically provisioned and released in response to user commands. These resources can be dynamically provisioned and reconfigured to adjust to variable load. Cloud computing can thus be considered as both the applications delivered as services over a publicly accessible network (e.g., the Internet, a cellular communication network) and the hardware and software in cloud provider data centers that provide those services.

A cloud system **102** can be formed as a number of regions, where a region is a separate geographical area in which the cloud provider clusters data centers. Each region can include two or more availability zones connected to one another via a private high-speed network, for example a fiber communication connection. An availability zone (also known as an availability domain, or simply a “zone”) refers to an isolated failure domain including one or more data center facilities with separate power, separate networking, and separate cooling from those in another availability zone. A data center

refers to a physical building or enclosure that houses and provides power and cooling to servers of the cloud provider network. Preferably, availability zones within a region are positioned far enough away from one other that the same natural disaster should not take more than one availability zone offline at the same time. Users **106** can connect to availability zones of the cloud provider network via a publicly accessible network (e.g., the Internet, a cellular communication network) by way of a transit center (TC). TCs can be considered as the primary backbone locations linking customers to the cloud provider network, and may be collocated at other network provider facilities (e.g., Internet service providers, telecommunications providers) and securely connected (e.g., via a VPN or direct connection) to the availability zones. Each region can operate two or more TCs for redundancy. Regions are connected to a global network which includes private networking infrastructure (e.g., fiber connections controlled by the cloud provider) connecting each region to at least one other region. The cloud system **102** may deliver content from points of presence outside of, but networked with, these regions by way of edge locations and regional edge cache servers. This compartmentalization and geographic distribution of computing hardware enables the cloud system **102** to provide low-latency resource access to customers on a global scale with a high degree of fault tolerance and stability.

With cloud computing, instead of buying, owning, and maintaining their own data centers and servers, organizations can acquire technology such as compute power, storage, databases, and other services on an as-needed basis. The cloud system **102** may provide on-demand, scalable computing services to users **106** through a network, for example allowing users **106** to have at their disposal scalable “virtual computing devices” via their use of the compute servers and block store servers. These virtual computing devices have attributes of a personal computing device including hardware (various types of processors, local memory, random access memory (“RAM”), hard-disk and/or solid state drive (“SSD”) storage), a choice of operating systems, networking capabilities, and pre-loaded application software. Each virtual computing device may also virtualize its console input and output (“I/O”) (e.g., keyboard, display, and mouse). This virtualization allows users **106** to connect to their virtual computing device using a computer application such as a browser, application programming interface, software development kit, or the like, in order to configure and use their virtual computing device just as they would a personal computing device. Unlike personal computing devices, which possess a fixed quantity of hardware resources available to the user **106**, the hardware associated with the virtual computing devices can be scaled up or down depending upon the resources the user **106** requires. Users **106** can choose to deploy their virtual computing systems to provide network-based services for their own use and/or for use by their users or client.

The cloud system **102** may implement various computing resources or services, which may include a virtual compute service, data processing service(s) (e.g., map reduce, data flow, and/or other large scale data processing techniques), data storage services (e.g., object storage services, block-based storage services, or data warehouse storage services) and/or any other type of network based services (which may include various other types of storage, processing, analysis, communication, event handling, visualization, and security services not illustrated). The resources required to support the operations of such services (e.g., compute and storage resources) may be provisioned in an account associated with

the cloud provider, in contrast to resources requested by users **106** of the cloud system **102**, which may be provisioned in user accounts.

The cloud system **102** may offer many different built-in services to the users **106** to help run their applications and services. For instance, the cloud system **102** may provide users **106** with use of VPCs, which are logically isolated sections of the cloud system **102** that serve as private virtual environments to which only permitted users **106** have access to use. Users **106** may have multiple VPCs, potentially spanning across different regions of the cloud system **102**. To help interconnect VPCs and other resources of the user’s computing infrastructure, the service provider may also offer many different built-in networking services. The VPCs may be used by the users **106** to create subnetworks, configure personalized route tables, choose IP addresses, monitor connections, screen network traffic, restrict access into the VPCs, and/or for other operations. As shown. The VPCs include or run computing resources, such as servers, virtual machines (VMs), containers, serverless functions, workloads, processes, etc. Generally, the computing resources are used to support or run applications or services of the users **106**.

A virtual private cloud (VPC) (also referred to as a virtual network (VNet), virtual private network, or virtual cloud network, in various implementations) is a custom-defined, virtual network within another network, such as a cloud provider network. A VPC can provide the foundational network layer for a cloud service, for example a compute cloud or an edge cloud, or for a customer application or workload that runs on the cloud. A VPC can be defined by at least its address space, internal structure (e.g., the computing resources that comprise the VPC, security groups), and transit paths.

A VPC is a virtual network dedicated to a particular customer account (or set of related customer accounts, such as different customer accounts belonging to the same business organization). A VPC is logically isolated from other virtual networks in the cloud. Customers can launch resources, such as compute instances, into a VPC. When creating a VPC, a customer can specify a range of IPv4 addresses for the VPC in the form of a Classless Inter-Domain Routing (CIDR) block. A VPC can span all of the availability zones in a particular region. After creating a VPC, a customer can add one or more subnets in each availability zone or edge location.

Users **106** may desire that the license manager service **104** provision licensed applications **120** and activate the licensed applications **120** with first-party and/or third-party providers of the software. The users **106** (e.g., network administrators, organization employees, etc.) may interact with the cloud system **102**, via their user account and/or one or more user portals or consoles **110** (e.g., web console, command line interface (CLI), application programming interface (API), etc.). The license manager service **104** may request permission from the user to manage the licenses used by resources in the VPCs of the user **106** via the console **110**. Further, the license manager service **104** may request that the user **106** specify whether the user **106** would like to activate licenses for applications running in peered VPCs of the configured VPC. The user **106** may provide answers to these questions, potentially along with other information used by the license manager service **104**, and submit an enrollment request **114** to the license manager service **104** over one or more networks **112**. In this way, users **106** may request that license manager service **104** perform different types of automated or semi-automated services such as launching, configuring,

licensing, and activating licenses for various licensed applications **120** for computing resources (e.g., servers, VMs, containers, etc.) in configured and peered VPCs.

To opt in for various services provided by the license manager service **104**, users **106** may define or provide a set of permissions, or roles, that the license manager service **104** may assume while in the user account **116**. The permissions define types of management operations that the license manager service **104** is permitted to take, and/or not permitted to take, as the license manager service **104** provides various automated management experiences. In some instances, the set of permissions created for the license manager service **104** may be referred to as a “service-linked role” that the license manager service **104** may assume when performing management operations in the user account **116**.

After the set of permissions are in the user account **116**, the license manager service **104** may request that the user **106** designate a configured VPC **118** to be configured for the licensed applications **120** to be launched into, and to communicate with activation servers **136**. Generally, the cloud system **102** may include one or more activation servers **136** that perform techniques for activating licenses of the licensed applications **120**. The configured VPC **118** that is selected for configuration will include at least one subnet in which an endpoint (or “gateway”) is provisioned that is configured to create a private connection with the activation servers **136** over which activation requests **126** are sent to activate the licensed applications **120**. The license manager service **104** may configure the configured VPC **118** by launching a VPC endpoint into a subnet of the VPC to establish the private connection with the activation servers **136**, applying security groups or policies that allow communications with the activation servers **136**, and configuring a directory service in the configured VPC **118** that manages permissions and credentials for activating the licensed applications **120**. In some examples, computing resources (e.g., servers, virtual machines, containers, etc.) on which licensed applications **120** are launched in the configured VPC **118** may be allowed to activate the licensed applications **120**.

The user account **116** may have additional VPCs in which computing resources are hosted, and some of those VPCs **122** may have peering connections with the configured VPC. As noted above, VPCs are virtual networks that are logically isolated from other virtual networks in the cloud system **102** in which computing resources are launched. A VPC peering connection **124** is a networking connection between two VPCs that enables traffic to be routed between the VPCs using private IP addresses (e.g., IPv4 and/or IPv6 addresses). Computing resources in peered VPCs **122** can communicate with each other over the peering connections **124** as if they are within the same virtual network. The VPC peering connections **124** can be created between VPCs registered to a single user account **116**, or between VPCs in different user accounts **116**, and the VPCs can be in different regions (also known as an inter-region VPC peering connection **124**).

As shown in console **110**, the user **106** may specify that the license manager service **104** is to activate licensed applications **120B** that are launched in peered VPCs **122** of the configured VPC **118**. The peered VPCs **122** may communicate the activation requests **126** from the licensed applications **120B** running in the peered VPCs **122** over the peered connection to the configured VPC **118**, and the activation requests **126** may then be forwarded towards an activation server **136** over the private connection between the configured VPC **118** and the activation server **136**.

As an example, the user account **116** may request that the license manager service **104**, or another service of the cloud

system **102**, provision a new instance of a licensed application **120B** in a peered VPC **122** of the configured VPC **118**. The license manager service **104** may launch a virtual resource (e.g., VM, container, etc.), or detect a launch of a virtual resource, on which the licensed application **120B** is installed. The license manager service **104** may configure the licensed application **120B** to send activation requests to an activation firewall **130** that is configured to allow or deny activation requests **126** destined to the activation server **136**. For instance, the licensed application **120B** may be configured to send activation requests **126** to a private DNS name that is assigned to the activation firewall **130**, and/or a load balancer that distributes activation requests to the activation firewall **130**. To trigger the activation process, the license manager service **104** may send a command to the licensed application **120B** to send the activation request **126** to the activation firewall **130** and begin the activation process. The licensed application **120B** running in the peered VPC **122** may send the activation request **126** over the peering connection **124** to the configured VPC **118**, and the activation request **126** may then be forwarded to the activation firewall **130** over the private connection.

The activation firewall **130** may be used to ensure that licensed applications **120** are only activated if activation of the licensed applications **120** are permitted by the license manager service **104**. In this way, the license manager service **104** may track usage of all licensed applications **120** and determine metrics indicating the usage of the licensed applications **120** for compliance reasons. For instance, there may be agreements in place regarding usage, such as payment or permitted use agreements, that indicate amounts of usage of licensed applications **120** that are permitted, or costs for running licensed applications **120** (e.g., cost per instance of a licensed application **120**, cost for how long a licensed application **120** is used, etc.).

To ensure that users **106** are unable to provision and activate license instances that are not tracked by the license manager service **104**, the activation firewall **130** may filter activation requests **126** received from virtual computing resources. Following the above example, the activation firewall **130** may receive the activation request **126** sent from the application **120B** on the virtual computing resource and determine the source of the activation request **126**. For instance, the activation firewall **130** (and/or load balancer) may extract metadata from the activation request **126** indicating that the virtual computing resource is a source of the activation request. The metadata may include a private IP address associated with virtual computing resource, a VPC ID of a VPC in which the virtual computing resource is provisioned, and an account identifier (ID) of the user account. The activation firewall **130** may send at least a tuple including the VPC ID of the peered VPC **122** and the private IP address (and potentially the user account ID) of the virtual computing resource to the license manager service **104** and/or a compute-management system to verify that the virtual computing resource is in a peered VPC **122** to the configured VPC **118** of the user account **116**.

The activation firewall **130** may be configured to drop, block, or otherwise refrain from sending activation requests **126** to the activation servers **136** upon determining that the activation requests **126** were received from virtual computing resources and/or applications that are running in non-peered VPCs **127** (e.g., denied requests **134**). Conversely, the activation firewall **130** may forward activation requests **126** to the activation server **136** upon determining that the activation requests **126** were received from virtual computing resources and/or applications **120** that are running in the

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configured VPC **118** or peered VPC **122** (e.g., allowed requests **132**). In this way, only licensed applications **120** managed by the license manager service **104** and launched in the configured VPC **118** or peered VPC **122** may be activated by activation servers **136**.

In some instances, there may be different activation servers **136** configured to activate different licensed applications **120**. The activation firewall **130** may determine a unique product code for the licensed applications **120** and store mappings between private DNS names for activation servers **136** and unique product codes, and forward activation requests **126** to the private DNS names that are mapped to the requests' unique product codes. In some instances, the activation firewall **130** and activation servers **136** may be in a license manager VPC **128** of the cloud system **102**. In other instances, however, the activation servers **136** may be at a remote location outside of the cloud system **102**.

The activation firewall **130** may help establish and maintain a connection open between the virtual computing resource and the activation server **136** to allow for activation of the application **120B** using an activation protocol. The connection may be a stateful Transport Control Protocol (TCP) connection that is used to communicate data for the activation protocol. In this way, the license manager service **104** may automate the process for provisioning and activating licensed applications **120** for use by users **106** of a cloud system **102**.

In some examples, the cloud system **102** may offer customer accounts that allow organizational users (e.g., enterprises, companies, groups of users, etc.) to create user accounts that are managed by administrators of the customer account. For instance, companies may register for one or more customer accounts under which individual user accounts are created for employees of the companies. The customer accounts allow for administrators of those accounts to allocate resources, group accounts, and apply governance policies to accounts or groups.

In addition to provisioning and activating licensed applications, the license manager service **104** may additionally assign user accounts of the customer account to have access to applications (licensed and unlicensed applications). To provide user accounts with access to licensed applications, the customer account may onboard a directory service with the license manager service **104**. For instance, an administrator of the customer account may provide the license manager service **104** with a directory ID of the customer account, and the manager service may locate and onboard the directory service using the directory ID. The administrator may provide input indicate what user account(s) in the customer account are to be provided access to use the licensed application **120**. The license manager service **104** may then, using the set of permissions, update the directory service to indicate the user account(s) **116** that are permitted to access the application **120** running in the customer account of the particular type. In this way, the applications **120** running on virtual resources are usable by the assigned user accounts **116** when the accounts present credentials (e.g., username and password, bio authenticator, etc.).

Thus, users **106** may have created user accounts **116** with the cloud system **102** to utilize the resources of the cloud system **102**. The users **106** may utilize their devices **108** to communicate over one or more networks **112** (e.g., WANs, PANs, LANs, etc.) with the cloud system **102**. The user devices **108** may comprise any type of computing device configured to communicate over network(s) **112**, such as mobile phones, tablets, laptop computers, desktop computers, televisions, servers, and/or any other type of computing

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device. The users **106** (e.g., network administrators, organization employees, etc.) may interact with the cloud system **102**, via their user account and/or one or more user portals or consoles **110** (e.g., web console, command line interface (CLI), application programming interface (API), etc.).

Generally, the license manager service **104**, and components thereof, may comprise software, firmware, and/or other logic that is supported one computing device, or across more computing devices in the cloud system **102**. Additionally, the license manager service **104** may comprise a system of other devices. The techniques described herein are generally described with respect to a service provider network, such as a cloud provider network or system. However, the techniques are generally applicable for any network, such as on-premises networks, hybrid networks, and so forth.

FIG. 2 illustrates an example user interface **200** through which a user **106** enrolls for use of a license manager service **104**. The user interface **200** provides various options for the user **106** to populate in order to enroll for license manager service **104** to provide the user **106** with a managed experience for managing licenses used by resources.

A service-linked-role (SLR) field **202** includes a request for the user **106** to grant the license manager service **104** with the ability to create a set of permissions, or an SLR, in their user account **116** to perform various operations (e.g., calling various APIs) for managing the licenses used by resources. The user **106** may select an option in the SLR field **202** to grant the license manager service **104** permission, and the license manager service **104** may create the SLR in the user account **116**.

A managed directory service field **204** may prompt the user **106** to select which directory service and products they would like to have managed. The user **106** may interact with a directory name and ID field **206** to select a particular directory service, and may also interact with a product field **208** to select product names and IDs for the products they would like to have managed by the license manager service **104** and directory service.

A VPC field **210** may prompt the user to choose a configured VPC **118** that is to receive additional configuration to enable the license manager service **104**. The user **106** may be able to filter or search all their VPCs and select a particular VPC to be the configured VPC **118** to enable the license manager service **104** (e.g., launching a VPC endpoint into a subnet of the VPC to establish the private connection with the activation servers **136**, applying security groups or policies that allow communications with the activation servers **136**, and configuring a directory service in the VPC **118** that manages permissions and credentials for activating the licensed applications **120**).

Additionally, the user interface **200** may include a peering option **214** that is selectable by the user **106** to enable the license manager service **104** to activate licenses for applications **120** running in peered VPCs **122** of the configured VPC **118**.

The user interface **200** may also include a subnets field **216** that allows the user **106** to select a subnet of the configured VPC **118** in which to launch a VPC endpoint used to establish the private connection. Further, the user interface **200** may include a security groups **218** field (e.g., firewall policy groupings) through which the user **106** may select one or more security groups that allow traffic inbound to the VPC endpoint that is in the subnet of the configured VPC **118**.

Once the user **106** has finished providing information needed to enroll for use of the license manager service **104**, the user **106** may select a configure option that causes the

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license manager service **104** to begin performing operations for automating the activation of licensed applications **120** in the configured VPC **118** and peered VPC **122**.

FIG. 3 illustrates a flow diagram **300** in which a license manager service **104** and components of a cloud system **102** activates a licensed application running in a peered VPC **122** for use by a user account **116**.

At **1**, the user **106** may access their console **110** and launch an instance. The request to launch an instance may be sent to the compute-management service **202**, and at **2**, the compute-management service **302** may boot the instance. For instance, the compute-management service **202** may allocate a launched and running instance for use by the particular user account and boot the instance in the user account. At **3**, the license manager service **104** may send an activation command that causes or triggers the licensed instance to send an activation request **126** (or the licensed instance is configured to send the activation request **126** on its own). At **4**, a VPC endpoint **305** may forward the activation request **126** over a private DNS link **306** that is connected to a network load balancer **308** associated with the activation firewall **130**. At **5**, the network load balancer **308** may prepend a proxy protocol header **310** to the activation request **126** that includes information that the activation server **136** would have been able to get by itself if the client was connecting directly to the server **136** rather than via a proxy. The information carried by the protocol may be the ones the activation server **136** would get using `getsockname()` and `getpeername()`, such as the address family (AF_INET for IPv4, AF_INET6 for IPv6, AF_UNIX, the socket protocol (SOCK_STREAM for TCP, SOCK_DGRAM for UDP), layer 3 source and destination addresses, and/or layer 4 source and destination ports if any.

At **6**, the network load balancer **308** may send the activation request **126** to the activation firewall **130**, and the activation firewall **130** may perform filtering on the activation request **126**, such as by applying one or more filtering rules. For instance, the activation firewall **130** may determine if the activation request **126** was sent from an application **120** running in a configured VPC **118**, and if not, determine if the activation request **126** was sent from an application **120** running in a peered VPC **122**. The activation firewall **130** may drop activation requests **126** that are not sent from a peered VPC **122** or configured VPC **118**. Additionally, or alternatively, the activation firewall **130** may determine if the activation request **126** was sent from an IP address on a “disallowed list” or an “allowed list” and allow or disallow the activation request **126** based on the IP address being on disallowed or allowed lists.

At **7**, the activation firewall **130** may perform a reverse lookup at **6** for the source of the activation request **126**. Generally, the activation firewall **130** may scan all proxy protocol headers **310** sent over the network load balancer **308** in order to reverse-lookup the request’s origination data. The data in the header **310** may include account ID, IP address, and VPC ID for the source of the activation request **126**. The activation firewall **130** may call a describe-instances API to fetch a product code associated with the originating instance **120** to verify that it is running in the peered VPC **122**. Once the activation firewall **130** validates request’s **126** origin as being authorized (e.g., from an instance managed by the license manager service **104**), the activation firewall **130** may identify a server **136** to which the activation request **126** is to be forwarded. The activation firewall may, at **8**, use a product code mapping **312** to identify an activation server **136** for the activation request **126**. The product code may be mapped to a private DNS of

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an activation server **136**, and the activation firewall **130** may, at **9**, send the activation request **126** to the identified activation server **136** via network load balances **314**.

FIG. 4 illustrates a system-architecture diagram of an example environment **400** in which a license manager service **104** uses roles in a user account **116** to determine a listing of peered VPCs for a configured VPC in the user account.

As shown, a customer account **402** (which may include multiple user accounts belonging to the customer) may include a configured VPC **118**, at least one peered VPC **122**, and at least one non-peered VPC **126**. Further, a license manager account **404**, which is provided to the license manager service **104**, may include or run a license manager VPC **128**. The license manager service **104** may run in the license manager VPC **128** to perform various operations as described herein.

A user **106** may interact with a console **110** and send one or more interactions **406** over the network(s) **112**, including a request to launch a licensed application **120** that is offered by an application store **408**. Further, the interactions **406** may include user input indicating that one or more customer-accepted roles **410** are permitted to be placed in the configured VPC **118** to allow the license manager service **104** perform one or more permitted operations.

To provide user accounts with access to applications **120**, the customer account **402** may onboard a directory service **412** defined in the configured VPC **118**. For instance, an administrator of the customer account **402** may provide the license manager service **104** with a directory ID of the customer account **402**, and the license manager service **104** may locate and onboard the directory service **412** using the directory ID. The administrator may provide input indicate what user account(s) in the customer account **402** are to be provided access to use the licensed applications **120**. The license manager service **104** may then, using the role(s) **410**, update the directory service **412** to indicate the user account (s) that are permitted to access the applications **120** running in the customer account **402** of the particular type. In this way, the applications **120** running on virtual resources are usable by the assigned user accounts when the accounts present credentials (e.g., username and password, bio authenticator, etc.). The directory service **412** may include various organizational units that map different entities to each other. In this example, “instances,” such as application instances and/or virtual resources, are mapped to user accounts that are allowed or authorized to access the application instances or virtual resources.

The license manager service **104** may utilize the customer-accepted role(s) **410** in order to perform various operations for activating licensed applications **120**. For instance, a licensed application **120** may send a request to the activation firewall **130** of the license manager service **104**, and the license manager service **104** may use the role(s) **410** to call a get-peered-VPCs API **414**. The get-peered-VPCs API **414** may return a peered-VPCs listing **416** for the activation firewall **130** to use to determine whether or not to activate the licensed application **120**.

FIG. 5 illustrates a flow diagram **500** in which a license manager service **104** activates a licensed application **120** in a peered VPC **122** of a configured VPC **118**.

At **502**, a user **106** enrolls with the license manager service **104** to have the license manager service **104** automate the management of activation of licensed applications **120** that run in a user account **116** of the user **106**. For

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instance, the user **106** may interact with a console **110** and send an enrollment request **114** to the license manager service **104**.

At **504**, the user **106** initiates a role creation with the license manager service **104** to create a role in the user account **116**. For instance, the user **106** may interact with a service-linked role field **202** to allow the license manager service **104** to create a customer-accepted role **410** in the user account **116**.

At **506**, the license manager service **104** may create the customer-accepted role **410** in a configured VPC **118** of the user account **116** that enables the license manager service **104** to perform operations for managing the activation of licensed applications **120**.

At **508**, the license manager service **104** may configure the configured VPC **508** to enable the license manager service **104** to activate licensed applications **120** in the configured VPC **508**, and also peered VPC(s) **122**. The license manager service **104** may configure the configured VPC **118** by launching a VPC endpoint into a subnet of the VPC to establish the private connection with the activation servers **136**, applying security groups or policies that allow communications with the activation servers **136**, and configuring a directory service in the configured VPC **118** that manages permissions and credentials for activating the licensed applications **120**.

At **510**, the compute-management service **302** may launch an instance of a licensed application **120** on behalf of the user **106** in the peered VPC **122** (e.g., automated launch, launch for scaling, user-requested launch, etc.).

At **512**, the licensed application **120** that was launched may send an activation request **126** to the activation firewall **130** in order to activate a license of the licensed application **120**.

At **514**, the activation firewall **130** may receive the activation request **126**, and determine a source of the activation request **126**. The activation firewall **130** may initially determine whether the licensed application **120** is in a configured VPC **118**.

In response to determining that the licensed application **120** that send the activation request **126** is not in the configured VPC **118**, the activation firewall **130** may determine, at **516**, that the licensed application is in the peered VPC **122**. At **518**, the activation firewall **130** may forward the activation request **126** to an activation server **136** that activates the licensed application **120**.

FIG. 6 illustrates a component diagram **600** of example components of a cloud system **102** that performs the techniques described herein. As illustrated, the cloud system **102** may include one or more hardware processors **602** (processors), one or more devices, configured to execute one or more stored instructions. The processor(s) **602** may comprise one or more cores. Further, the cloud system **102** may include one or more network interfaces **604** configured to provide communications between the cloud system **102** and other devices, such as the user device(s), and/or other systems or devices in the cloud system **102** and/or remote from the cloud system **102**. The network interfaces **604** may include devices configured to couple to personal area networks (PANs), wired and wireless local area networks (LANs), wired and wireless wide area networks (WANs), and so forth.

The cloud system **102** may also include computer-readable media **606** that stores various executable components (e.g., software-based components, firmware-based components, etc.). In addition to various components discussed in the previous figures, the computer-readable-media **606** may

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further store components to implement functionality described herein. While not illustrated, the computer-readable media **606** may store one or more operating systems utilized to control the operation of the one or more devices that comprise the cloud system **102**. According to one embodiment, the operating system comprises the LINUX operating system. According to another embodiment, the operating system(s) comprise the WINDOWS SERVER operating system from MICROSOFT Corporation of Redmond, Washington. According to further embodiments, the operating system(s) can comprise the UNIX operating system or one of its variants. It should be appreciated that other operating systems can also be utilized.

Additionally, the cloud system **102** may include a data store, or storage **608**, which may comprise one, or multiple, repositories or other storage locations for persistently storing and managing collections of data such as databases, simple files, binary, and/or any other data. The storage **608** may include one or more storage locations that may be managed by one or more database management systems.

The computer-readable media **606** may store services, or components, of the cloud system **102** as described herein. Further, the computer-readable media **606** may store an application-management service **610** configured to manage the applications are offered for use in the cloud system **102** (e.g., in the application store **408**). Further, the computer-readable media **606** may store and execute a metering service **612** to track the number of instances of an application running for compliance and cost reasons. Additionally, the computer-readable media **606** may store and execute an audit service **614** that audits usage of applications **120** to ensure that users **106** are compliant with rules for using the applications **120** (e.g., identifying untracked application instances and onboarding them).

To utilize the services provided by the cloud system **102**, the users **106** may register for accounts **116** with the cloud system **102**. For instance, users **106** may utilize a user device to interact with an identity and access management (IAM) component **616** that allows the users **106** to create user accounts **116** with the cloud system **102**. Generally, the IAM component **616** may enable the users **106** to manage their network infrastructures remotely, and view analytics provided by the license manager service **104**. Generally, the different user accounts **116** can assume different roles, or sets or permissions/credentials, that allow users **106** to perform different actions, and be restricted from performing some actions. In some instances, a same organization may have multiple accounts that have different network infrastructures.

The computer-readable media **606** may be used to store and retrieve information, such as program modules, data structures, or other data. It should be appreciated by those skilled in the art that computer-readable storage media is any available media that provides for the non-transitory storage of data and that can be accessed by the cloud system **102**. In some examples, the operations performed by the cloud system **102**, and or any components included therein, may be supported by one or more server devices. Stated otherwise, some or all of the operations performed by the cloud system **102**, and or any components included therein, may be performed by one or more computer devices operating in a cloud-based arrangement.

By way of example, and not limitation, computer-readable storage media **606** can include volatile and non-volatile, removable and non-removable media implemented in any method or technology. Computer-readable storage media includes, but is not limited to, RAM, ROM, erasable pro-

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grammable ROM (“EPROM”), electrically-erasable programmable ROM (“EEPROM”), flash memory or other solid-state memory technology, compact disc ROM (“CD-ROM”), digital versatile disk (“DVD”), high definition DVD (“HD-DVD”), BLU-RAY, or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information in a non-transitory fashion.

FIGS. 7-9 illustrate flow diagrams of example methods 700, 800, and 900, that illustrate aspects of the functions performed at least partly by the cloud system 102 as described in this disclosure. The logical operations described herein with respect to FIGS. 7-9 may be implemented (1) as a sequence of computer-implemented acts or program modules running on a computing system and/or (2) as interconnected machine logic circuits or circuit modules within the computing system.

The implementation of the various components described herein is a matter of choice dependent on the performance and other requirements of the computing system. Accordingly, the logical operations described herein are referred to variously as operations, structural devices, acts, or modules. These operations, structural devices, acts, and modules can be implemented in software, in firmware, in special purpose digital logic, and any combination thereof. It should also be appreciated that more or fewer operations might be performed than shown in the FIGS. 7-9 and described herein. These operations can also be performed in parallel, or in a different order than those described herein. Some or all of these operations can also be performed by components other than those specifically identified. Although the techniques described in this disclosure is with reference to specific components, in other examples, the techniques may be implemented by less components, more components, different components, or any configuration of components.

FIG. 7 illustrate a flow diagram of an example process 700 for a license manager service 104 to configure a VPC, determine peered VPCs for the configured VPC, and forward an activation request sent from a licensed application in the peered VPC to an activation server.

At 702, the license manager service may receive an enrollment request for the license manager to activate licensed application instances on behalf of a user account registered with the cloud system. For instance, a user 106 may submit an enrollment request 114 to the license manager service 104 to active licensed applications 120 on behalf of the user account 116 that is registered with the cloud system 102.

At 704, the license manager service may receive first input indicating that a first virtual private cloud (VPC), from a group of VPCs associated with the user account, to configure such that virtual computing resources launched in the first VPC are permitted to activate the licensed application instances. In some instances, the first input may be received via a VPC field 210 that prompts the user 106 to choose a configured VPC 118 that is to receive additional configuration to enable the license manager service 104.

At 706, the license manager service may provision a virtual computing resource on behalf of the user account to run a licensed application instance. For instance, the compute-management service 302 may provision a virtual computing resource on behalf of the user account 116 to run a licensed application 120 instance.

At 708, the license manager service may configure the licensed application instance to send an activation request to an activation firewall that allows or denies activation

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requests. In some instances, the licensed application 120 instance may send an activation request 126 in response to receiving a trigger from the license manager service 104, and in other examples, the licensed application 120 instance may send an activation request 126 upon being installed or launched.

At 710, the license manager service 104 may receive, at the activation firewall 130, the activation request 126 sent from the virtual computing resource to activate the licensed application 120 instance.

At 712, the license manager service 104 may determine that the virtual computing resource from which the activation request was sent is provisioned in a second VPC of the group of VPCs. For instance, the activation firewall 130 may extract a tuple from the activation request 126 where the tuple includes a VPC identifier (ID) of the second VPC and a private Internet Protocol (IP) address of the virtual computing resource, and identify the second VPC using the VPC ID.

At 714, a license manager service 104 may determine that the second VPC has a peering connection with the first VPC. For instance, the license manager service 104 may verify, using the tuple and a listing of peered VPCs 122, that the virtual computing resource is provisioned in the second VPC (or peered VPC 122) having the peered connection 124 with the first VPC (or configured VPC 118).

At 716, the license manager service 104 may, based at least in part on the second VPC having the peering connection 124 with the first VPC, forwarding the activation request 126 from the activation firewall 130 and to an activation server 136 that activates licensed applications 120 for use. The activation server 136 may then communicate with the licensed application 120 instance to activate the licensed application.

FIG. 8 illustrate a flow diagram of an example process 800 for a license manager service 104 to receive an activation request 126 sent from a licensed application 120, determine that the licensed application 120 is in a peered VPC 122 of a configured VPC 118, and forward the activation request 126 to an activation server 136.

At 802, the license manager service 104 may establish a first connection between a first virtual network and the activation firewall such that computing resources in the first virtual network are configured to send activation requests to the activation firewall. For instance, a configured VPC 118 may use a VPC endpoint 305 to create a private DNS link 306 to an activation firewall 130. Generally, the first and second virtual networks may be overlay networks that provide isolated portions of a cloud system 102 to different users or customers. Examples of the overlay network may be VPCs, virtual networks (or VNs), and/or other virtual overlay networks.

At 804, the license manager service 104 may receive, at the activation firewall 130, an activation request 126 from a particular computing resource to activate a licensed application 120 instance.

At 806, the license manager service 104 may determine that the particular computing resource is running in a second virtual network. For instance, the activation firewall 130 may extract a tuple from the activation request 126 where the tuple includes a VPC ID of the second VPC and a private IP address of the virtual computing resource, and identify the second VPC using the VPC ID.

At 808, a license manager service may determine that the second virtual network is peered with the first virtual network in that the second virtual network has a second connection established with the first virtual network. For

instance, the license manager service **104** may determine that the second virtual network (e.g., peered VPC **122**) has a peering connection **124** with the first virtual network (e.g., configured VPC **118**).

At **810**, the license manager service **104** may, based at least in part on the second virtual network being peered with the first virtual network, forward the activation request **126** to the activation server **136** to activate the licensed application **120** instance. For example, the license manager service **104** may determine, at the activation firewall **136**, a unique product code that is assigned to the licensed application instance **120**, and identify, using a mapping stored at the activation firewall **130**, a private DNS name that is mapped to the unique product code where the private DNS name is associated with the activation server **136**. In such examples, the activation request **126** is forwarded to the private DNS name.

FIG. **9** illustrate a flow diagram of an example process **900** for an activation firewall **130** to filter activation requests **126** from licensed applications **120**.

At **902**, the activation firewall may establish a first connection with a first virtual network in a cloud system. For instance, the activation firewall **130** may establish the private DNS link **306** with the configured VPC **118** in the cloud system **102**.

At **904**, the activation firewall may receive an activation request to activate a licensed application instance running on a computing resource. For instance, an activation request **126** may be received from a licensed application **120** running in a VPC of a user account **116**.

At **906**, the activation firewall may determine that the computing resource that sent the activation request is running in a second virtual network in the cloud system. For instance, the activation firewall **130** may extract a tuple from the activation request **126** where the tuple includes a VPC ID of the second virtual network and a private IP address of the computing resource, and identify the second virtual network using the VPC ID.

At **908**, the activation firewall may determine whether the second virtual network is peered with the first virtual network. For instance, the activation firewall **130** may determine whether the second virtual network (e.g., second VPC) has a peering connection **124** established with the first virtual network (e.g., first VPC).

At **910**, in response to determining that the second virtual network is peered to the first virtual network, the activation firewall **130** may forward the activation request **126** to an activation server. Alternatively, in response to determining that the second virtual network is not peered to the first virtual network, activation firewall **130** may drop the activation request **126** at **912**.

FIG. **10** is a system and network diagram **1000** that shows an illustrative operating environment that includes data centers **1004** in one or more regions **1006** of a cloud system **102** that can be configured to implement aspects of the functionality described herein. The cloud system **102** can provide computing resources, like VM instances and storage, on a permanent or an as-needed basis. Among other types of functionality, the computing resources provided by the cloud system **102** may be utilized to implement the various services described above. As also discussed above, the computing resources provided by the cloud system **102** can include various types of computing resources, such as data processing resources like VM instances, data storage resources, networking resources, data communication resources, network services, and the like.

Each type of computing resource provided by the cloud system **102** can be general-purpose or can be available in a number of specific configurations. For example, data processing resources can be available as physical computers or VM instances in a number of different configurations. The VM instances can be configured to execute applications, including web servers, application servers, media servers, database servers, gaming applications, some or all of the network services described above, and/or other types of programs. Data storage resources can include file storage devices, block storage devices, and the like. The cloud system **102** can also be configured to provide other types of computing resources not mentioned specifically herein.

The computing resources provided by the cloud system **102** may be enabled in one embodiment by one or more data centers **1004A-1004N** (which might be referred to herein singularly as “a data center **1004**” or in the plural as “the data centers **1004**”). The data centers **1004** are facilities utilized to house and operate computer systems and associated components. The data centers **1004** typically include redundant and backup power, communications, cooling, and security systems. The data centers **1004** can also be located in geographically disparate locations, or regions **1006**. One illustrative embodiment for a data center **1004** that can be utilized to implement the technologies disclosed herein will be described below with regard to FIG. **11**.

The users **106**, such as administrators **106**, of the user devices **108** that utilize the cloud system **102** may access the computing resources provided by the cloud system **102** over any wired and/or wireless network(s) **112**, which can be a wide area communication network (“WAN”), such as the Internet, an intranet or an Internet service provider (“ISP”) network or a combination of such networks. For example, and without limitation, a user device operated by a user **106** of the cloud system **102** may be utilized to access the cloud system **102** by way of the network(s) **112**. It should be appreciated that a local-area network (“LAN”), the Internet, or any other networking topology known in the art that connects the data centers **1004** to remote clients and other users can be utilized. It should also be appreciated that combinations of such networks can also be utilized.

FIG. **11** is a computing system diagram **1100** that illustrates one configuration for a data center **1004** that implements aspects of the technologies disclosed herein. The example data center **1004** shown in FIG. **11** includes several server computers **1102A-1102F** (which might be referred to herein singularly as “a server computer **1102**” or in the plural as “the server computers **1102**”) for providing computing resources **1104A-1104E**. In some examples, the resources **1104** and/or server computers **1102** may include, be included in, or correspond to, the computing devices described herein.

The server computers **1102** can be standard tower, rack-mount, or blade server computers configured appropriately for providing the computing resources described herein (illustrated in FIG. **11** as the computing resources **1104A-1104E**). As mentioned above, the computing resources provided by the cloud system **102** can be data processing resources such as VM instances or hardware computing systems, database clusters, computing clusters, storage clusters, data storage resources, database resources, networking resources, and others. Some of the servers **1102** can also be configured to execute a resource manager **1106** capable of instantiating and/or managing the computing resources. In the case of VM instances, for example, the resource manager **1106** can be a hypervisor or another type of program configured to enable the execution of multiple VM instances

on a single server computer **1102**. Server computers **1102** in the data center **1004** can also be configured to provide network services and other types of services.

In the example data center **1004** shown in FIG. **11**, an appropriate LAN **1108** is also utilized to interconnect the server computers **1102A-1102F**. It should be appreciated that the configuration and network topology described herein has been greatly simplified and that many more computing systems, software components, networks, and networking devices can be utilized to interconnect the various computing systems disclosed herein and to provide the functionality described above. Appropriate load balancing devices or other types of network infrastructure components can also be utilized for balancing a load between each of the data centers **1104A-1104N**, between each of the server computers **1102A-1102F** in each data center **1004**, and, potentially, between computing resources in each of the server computers **1102**. It should be appreciated that the configuration of the data center **1004** described with reference to FIG. **11** is merely illustrative and that other implementations can be utilized.

FIG. **12** shows an example computer architecture for a computer **1200** capable of executing program components for implementing the functionality described above. The computer architecture shown in FIG. **12** illustrates a conventional server computer, workstation, desktop computer, laptop, tablet, network appliance, e-reader, smartphone, or other computing device, and can be utilized to execute any of the software components presented herein.

The computer **1200** includes a baseboard **1202**, or “motherboard,” which is a printed circuit board to which a multitude of components or devices can be connected by way of a system bus or other electrical communication paths. In one illustrative configuration, one or more central processing units (“CPUs”) **1204** operate in conjunction with a chipset **1206**. The CPUs **1204** can be standard programmable processors that perform arithmetic and logical operations necessary for the operation of the computer **1200**.

The CPUs **1204** perform operations by transitioning from one discrete, physical state to the next through the manipulation of switching elements that differentiate between and change these states. Switching elements generally include electronic circuits that maintain one of two binary states, such as flip-flops, and electronic circuits that provide an output state based on the logical combination of the states of one or more other switching elements, such as logic gates. These basic switching elements can be combined to create more complex logic circuits, including registers, adders-subtractors, arithmetic logic units, floating-point units, and the like.

The chipset **1206** provides an interface between the CPUs **1204** and the remainder of the components and devices on the baseboard **1202**. The chipset **1206** can provide an interface to a RAM **1208**, used as the main memory in the computer **1200**. The chipset **1206** can further provide an interface to a computer-readable storage medium such as a read-only memory (“ROM”) **1210** or non-volatile RAM (“NVRAM”) for storing basic routines that help to startup the computer **1200** and to transfer information between the various components and devices. The ROM **1210** or NVRAM can also store other software components necessary for the operation of the computer **1200** in accordance with the configurations described herein.

The computer **1200** can operate in a networked environment using logical connections to remote computing devices and computer systems through a network, such as the network **112**. The chipset **1206** can include functionality for

providing network connectivity through a network interface controller (NIC) **1212**, such as a gigabit Ethernet adapter. The NIC **1212** is capable of connecting the computer **1200** to other computing devices over the local area network **1118** (or **112**). It should be appreciated that multiple NICs **1212** can be present in the computer **1200**, connecting the computer to other types of networks and remote computer systems.

The computer **1200** can include storage **1214** (e.g., disk) that provides non-volatile storage for the computer. The storage **1214** can consist of one or more physical storage units. The storage **1214** can store information by altering the magnetic characteristics of a particular location within a magnetic disk drive unit, the reflective or refractive characteristics of a particular location in an optical storage unit, or the electrical characteristics of a particular capacitor, transistor, or other discrete component in a solid-state storage unit. Other transformations of physical media are possible without departing from the scope and spirit of the present description, with the foregoing examples provided only to facilitate this description. The computer **1200** can further read information from the storage **1214** by detecting the physical states or characteristics of one or more particular locations within the physical storage units.

In addition to the storage **1214** described above, the computer **1200** can have access to other computer-readable storage media to store and retrieve information, such as program modules, data structures, or other data. It should be appreciated by those skilled in the art that computer-readable storage media is any available media that provides for the non-transitory storage of data and that can be accessed by the computer **1200**. In some examples, the operations performed by the cloud system **102**, and or any components included therein, may be supported by one or more devices similar to computer **1200**. Stated otherwise, some or all of the operations performed by the cloud system **102**, and or any components included therein, may be performed by one or more computer devices **1200** operating in a network-based arrangement.

By way of example, and not limitation, computer-readable storage media can include volatile and non-volatile, removable and non-removable media implemented in any method or technology. Computer-readable storage media includes, but is not limited to, RAM, ROM, erasable programmable ROM (“EPROM”), electrically-erasable programmable ROM (“EEPROM”), flash memory or other solid-state memory technology, compact disc ROM (“CD-ROM”), digital versatile disk (“DVD”), high definition DVD (“HD-DVD”), BLU-RAY, or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information in a non-transitory fashion.

The storage **1214** can store an operating system utilized to control the operation of the computer **1200**. According to one embodiment, the operating system comprises the LINUX operating system. According to another embodiment, the operating system comprises the WINDOWS SERVER operating system from MICROSOFT Corporation of Redmond, Washington. According to further embodiments, the operating system can comprise the UNIX operating system or one of its variants. It should be appreciated that other operating systems can also be utilized. The storage **1214** can store other system or application programs and data utilized by the computer **1200**.

In one embodiment, the storage **1214**, RAM **1208**, ROM **1210**, and/or other computer-readable storage media may be

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encoded with computer-executable instructions which, when loaded into the computer **1200**, transform the computer from a general-purpose computing system into a special-purpose computer capable of implementing the embodiments described herein. These computer-executable instructions transform the computer **1200** by specifying how the CPUs **1204** transition between states, as described above. According to one embodiment, the computer **1200** has access to computer-readable storage media storing computer-executable instructions which, when executed by the computer **1200**, perform the various techniques described above. The computer **1200** can also include computer-readable storage media having instructions stored thereupon for performing any of the other computer-implemented operations described herein.

The computer **1200** can also include one or more input/output controllers **1216** for receiving and processing input from a number of input devices, such as a keyboard, a mouse, a touchpad, a touch screen, an electronic stylus, or other type of input device. Similarly, an input/output controller **1216** can provide output to a display, such as a computer monitor, a flat-panel display, a digital projector, a printer, or other type of output device. It will be appreciated that the computer **1200** might not include all of the components shown in FIG. **12**, can include other components that are not explicitly shown in FIG. **12**, or might utilize an architecture completely different than that shown in FIG. **12**.

While the foregoing invention is described with respect to the specific examples, it is to be understood that the scope of the invention is not limited to these specific examples. Since other modifications and changes varied to fit particular operating requirements and environments will be apparent to those skilled in the art, the invention is not considered limited to the example chosen for purposes of disclosure, and covers all changes and modifications which do not constitute departures from the true spirit and scope of this invention.

Although the application describes embodiments having specific structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are merely illustrative some embodiments that fall within the scope of the claims of the application.

What is claimed is:

1. A method performed by a license manager service of a cloud system to activate licensed applications that are running in the cloud system, the method comprising:

receiving an enrollment request for the license manager service to activate licensed application instances on behalf of a user account registered with the cloud system;

receiving first input indicating that a first virtual private cloud (VPC), from a group of VPCs associated with the user account, to configure such that virtual computing resources launched in the first VPC are permitted to activate the licensed application instances;

provisioning a virtual computing resource on behalf of the user account to run a licensed application instance;

configuring the licensed application instance to send an activation request to an activation firewall that allows or denies activation requests;

receiving, at the activation firewall, the activation request sent from the virtual computing resource to activate the licensed application instance;

determining that the virtual computing resource from which the activation request was sent is provisioned in a second VPC of the group of VPCs;

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determining that the second VPC has a peering connection with the first VPC; and

based at least in part on the second VPC having the peering connection with the first VPC, forwarding the activation request from the activation firewall and to an activation server that activates licensed applications for use.

2. The method of claim 1, further comprising:

receiving consent to create a set of permissions in the user account that allows the license manager service to perform various operations; and

using the set of permissions, obtaining a listing of peered VPCs that have respective peering connections with the first VPC,

wherein determining that the second VPC has the peering connection with the first VPC includes identifying the second VPC from the listing of peered VPCs.

3. The method of claim 1, further comprising:

receiving a listing of peered VPCs that have respective peering connections with the first VPC;

extracting, by the activation firewall, a tuple from the activation request, the tuple including a VPC identifier (ID) of the second VPC and a private Internet Protocol (IP) address of the virtual computing resource; and

verifying, using the tuple and the listing of peered VPCs, that the virtual computing resource is provisioned in the second VPC having the peered connection with the first VPC.

4. The method of claim 1, further comprising:

provisioning the activation firewall and the activation server in an activation virtual private cloud (VPC) of the cloud system;

providing a VPC endpoint service associated with the activation VPC that accepts activation request on behalf of the activation firewall;

establishing a private connection between the VPC endpoint service and a user VPC endpoint associated with the first VPC,

wherein the activation request is sent from the second VPC to the activation firewall via the peering connection and the private connection.

5. A cloud system that runs a license manager service, the cloud system comprising:

an activation server that activates licensed applications;

an activation firewall that allows or denies activation requests from being sent to the activation server;

one or more processors; and

one or more computer-readable media storing computer-executable instructions that, when executed by the one or more processors, cause the license manager service to perform operations comprising:

establishing a first connection between a first virtual network and the activation firewall such that computing resources in the first virtual network are configured to send activation requests to the activation firewall;

receiving, at the activation firewall, an activation request from a particular computing resource to activate a licensed application instance;

determining that the particular computing resource is running in a second virtual network;

determining that the second virtual network is peered with the first virtual network in that the second virtual network has a second connection established with the first virtual network; and

based at least in part on the second virtual network being peered with the first virtual network, forward-

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ing the activation request to the activation server to activate the licensed application instance.

6. The cloud system of claim 5, wherein the activation request is sent from the second virtual network to the first virtual network via the second connection and from the first virtual network to the activation server via the first connection, the operations further comprising:

receiving, at the licensed application instance and via the first and second connection, an activation response sent from the activation server that activates the licensed application instance.

7. The cloud system of claim 5, the operations further comprising:

creating a set of permissions in a user account associated with the first virtual network that allows the license manager service to perform various operations; and using the set of permissions, obtaining a listing of peered virtual networks that have respective peering connections with the first virtual network,

wherein determining that the second virtual network is peered with the first virtual network includes identifying the second virtual network from the listing of peered virtual networks.

8. The cloud system of claim 5, the operations further comprising:

extracting, by the activation firewall, metadata from the activation request, wherein the metadata includes:

an Internet Protocol (IP) address associated with particular computing resource; and

a network ID of the second virtual network in which the particular computing resource is running;

verifying, using the metadata, that the particular computing resource is running in the second virtual network that is peered with the first virtual network.

9. The cloud system of claim 5, the operations further comprising:

receiving an enrollment request for the license manager service to activate licensed application instances on behalf of a user account registered with the cloud system;

prompting the user account for permission to create a set of permissions in the user account that allows the license manager service to perform operations;

receiving approval from the user account to create the set of permissions; and

creating the set of permissions in the user account associated with the first virtual network that allows the license manager service to perform the operations.

10. The cloud system of claim 5, wherein the first and second virtual networks are associated with a user account, the operations further comprising:

identifying, from the activation request, an Internet Protocol (IP) address associated with particular computing resource from the activation request;

determining that the IP address is also assigned to another computing resource in a third virtual network;

extracting, from the activation request, an identifier associated with at least one of the particular computing resource or the licensed application instance; and

verifying, using the identifier, that the particular computing resource is running in the second virtual network that is peered with the first virtual network.

11. The cloud system of claim 5, the operations further comprising:

provisioning the activation firewall and the activation server in a third virtual network of the cloud system;

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provisioning a first endpoint in the first virtual network; and

providing a second endpoint associated with the third virtual network that accepts activation request on behalf of the activation firewall;

wherein the first connection is established using the first and second endpoints.

12. The cloud system of claim 5, wherein the activation firewall is a containerized service with a plurality of firewall instances that each filter activation requests, the operations further comprising:

a network load balancer that receives and distributes the activation requests amongst the plurality of firewall instances.

13. The cloud system of claim 12, the operations further comprising:

determining, at the network load balancer, connection information associated with a connection used by the virtual computing instance to send the activation request; and

prepending, using a proxy protocol, the connection information in a packer header associated with the activation request such that the connection information is communicated to the activation server along with the activation request.

14. The cloud system of claim 5, the operations further comprising:

determining, at the activation firewall, a unique product code that is assigned to the licensed application instance; and

identifying, using a mapping stored at the activation firewall, a private Domain Name System (DNS) name that is mapped to the unique product code, the private DNS name being associated with the activation server, wherein the activation request is forwarded to the private DNS name.

15. A method performed at least partly by an activation firewall in a cloud system, comprising:

establishing a first connection between the activation firewall and a first virtual network in the cloud system; receiving, at the activation firewall, an activation request to activate a licensed application instance running on a computing resource;

determining, at the activation firewall, that the computing resource that sent the activation request is running in a second virtual network in the cloud system;

determining whether the second virtual network is peered with the first virtual network; and

in response to determining that the second virtual network is peered to the first virtual network, forwarding the activation request to an activation server; or

in response to determining that the second virtual network is not peered to the first virtual network, dropping the activation request.

16. The method of claim 15, wherein the activation request is sent from the second virtual network to the first virtual network via a second connection and from the first virtual network to the activation server via the first connection, further comprising:

determining that the second virtual network is peered to the first virtual network;

forwarding the activation request to an activation server; and

receiving, at the licensed application instance and via the first and second connection, an activation response sent from the activation server that activates the licensed application instance.

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17. The method of claim 15, further comprising:
 extracting, by the activation firewall, metadata from the
 activation request, wherein the metadata includes:
 an Internet Protocol (IP) address associated with particular
 computing resource; and
 a network ID of the second virtual network in which the
 particular computing resource is running;
 verifying, using the metadata, that the particular computing
 resource is running in the second virtual network
 that is peered with the first virtual network.

18. The method of claim 17, wherein the activation
 firewall is a containerized service with a plurality of firewall
 instances that each filter activation requests, further comprising:
 receiving, from a network load balancer, the activation
 requests amongst the plurality of firewall instances; and
 allowing or disallowing the activation requests by the
 plurality of firewall instances.

19. The method of claim 15, wherein the first and second
 virtual networks are associated with a user account, further
 comprising:

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identifying, from the activation request, an Internet Protocol (IP)
 address associated with particular computing resource from the
 activation request;
 determining that the IP address is also assigned to another
 computing resource in a third virtual network;
 extracting, from the activation request, an identifier associated
 with at least one of the particular computing resource or the
 licensed application instance; and
 verifying, using the identifier, that the particular computing
 resource is running in the second virtual network that is
 peered with the first virtual network.

20. The method of claim 15, further comprising:
 determining, at the activation firewall, a unique product
 code that is assigned to the licensed application instance; and
 identifying, using a mapping stored at the activation
 firewall, a private Domain Name System (DNS) name that is
 mapped to the unique product code, the private DNS name
 being associated with the activation server, wherein the
 activation request is forwarded to the private DNS name.

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